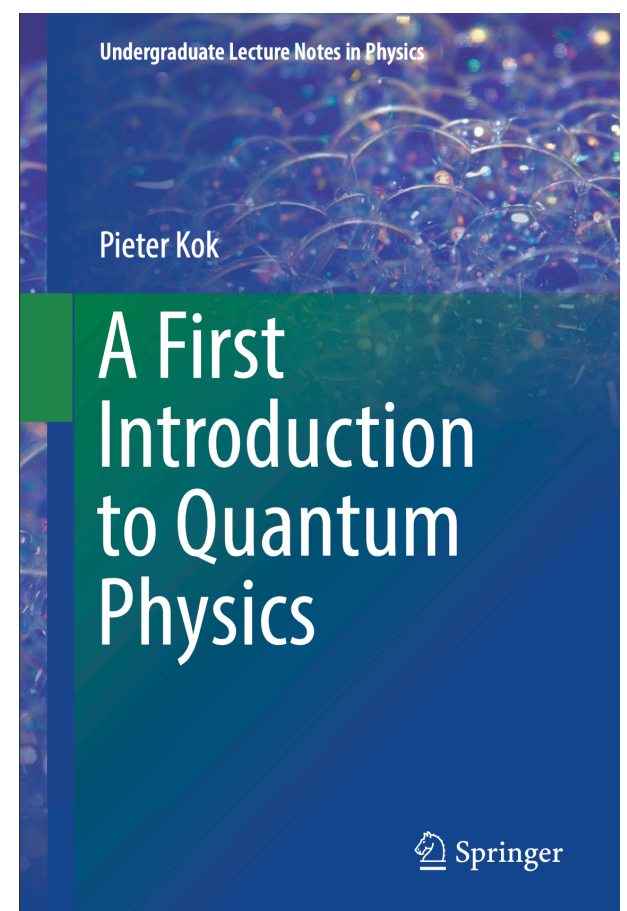
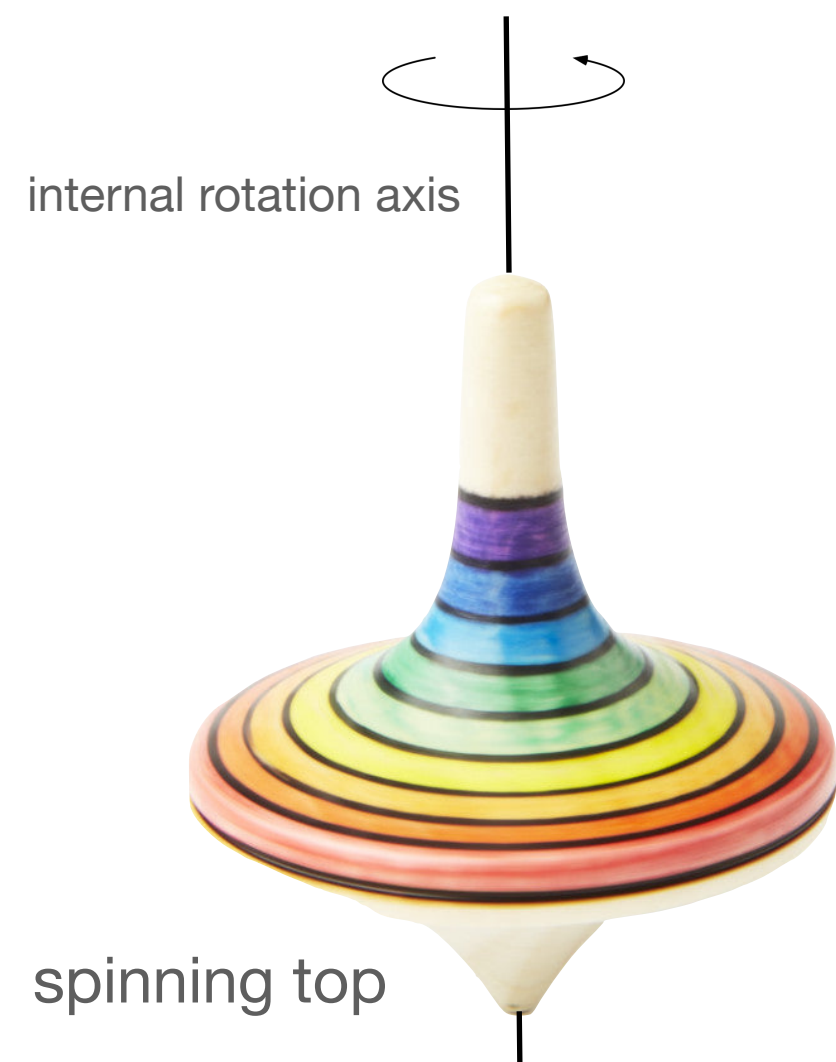


Introduction

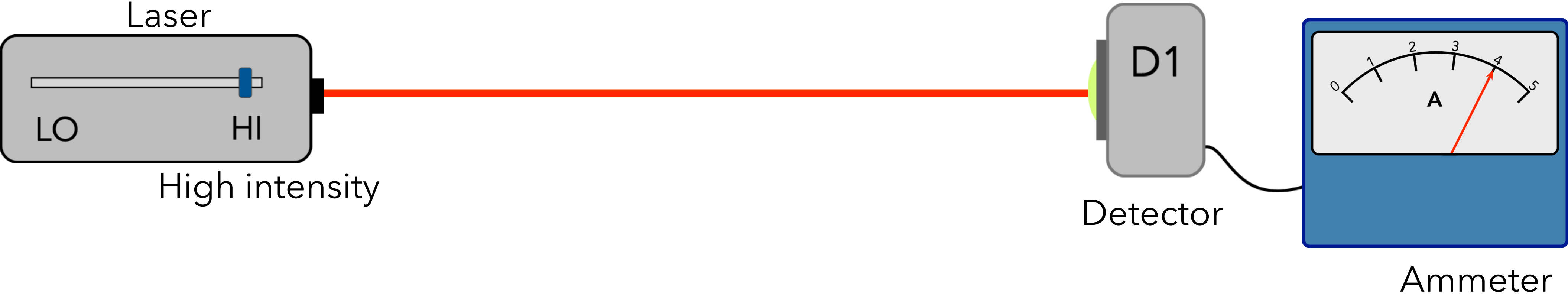
- Three laser experiments: description of the outcomes
- A basic mathematical description of a quantum state
- Stern-Gerlach: electron spin
- Expectation values: measurements



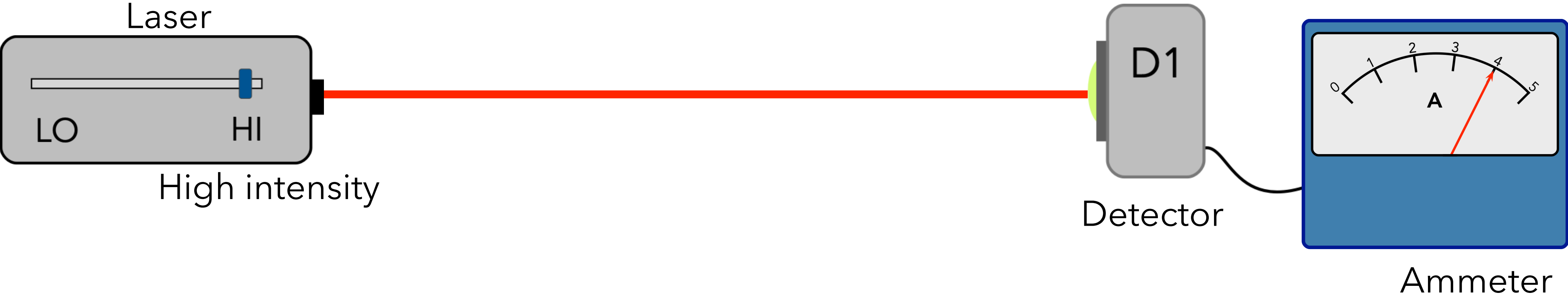
Experiment 1



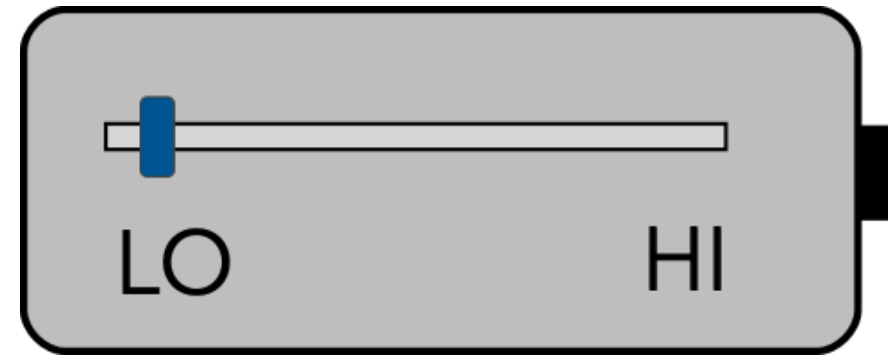
Experiment 1



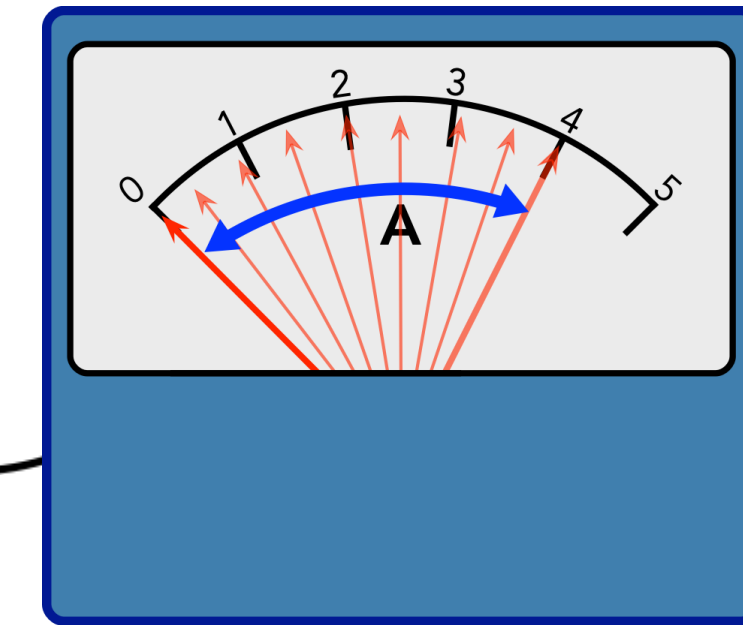
Experiment 1



Experiment 1

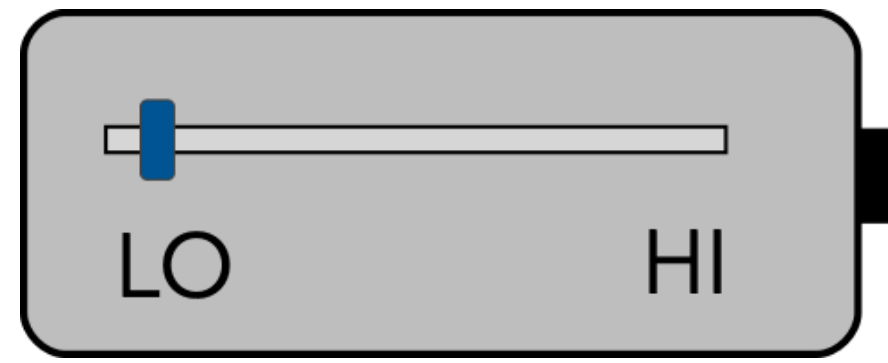


Low intensity



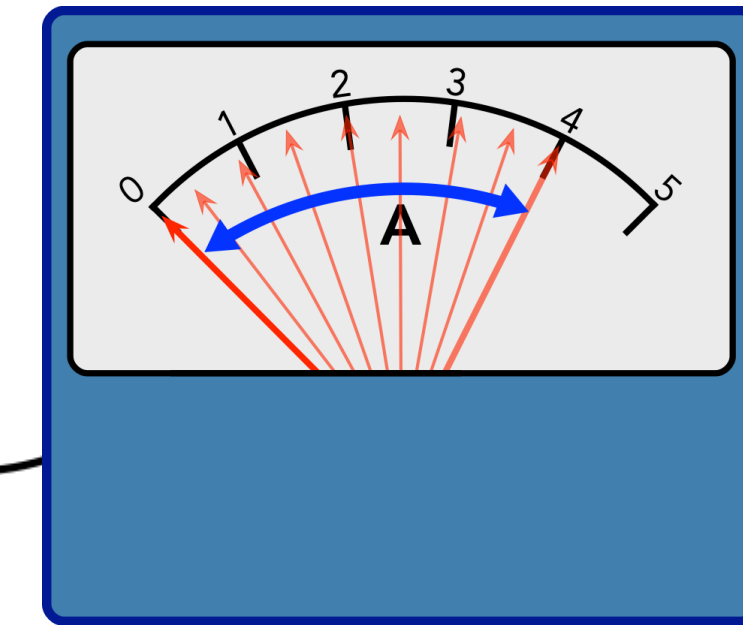
short pulses

Experiment 1



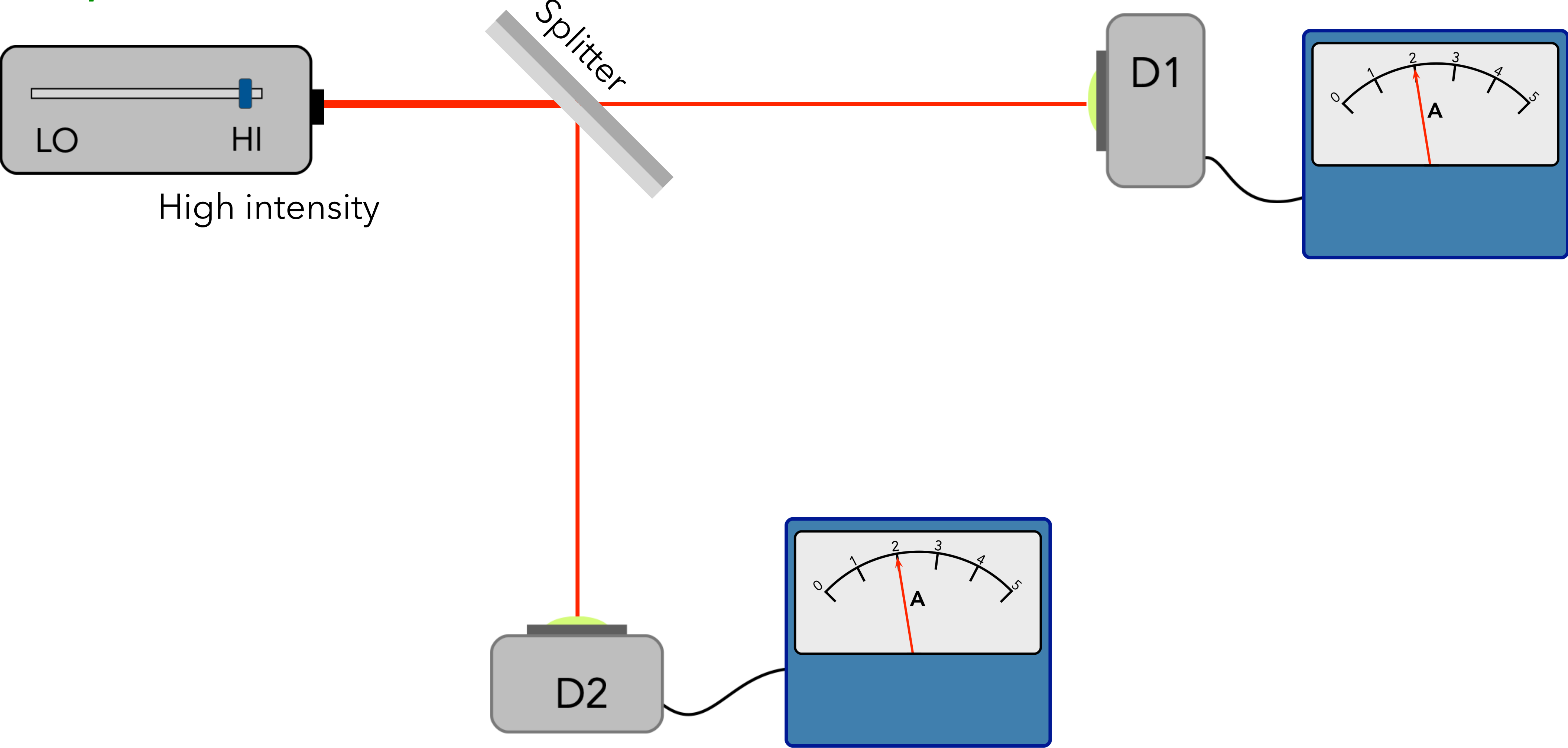
Low intensity

particles ?

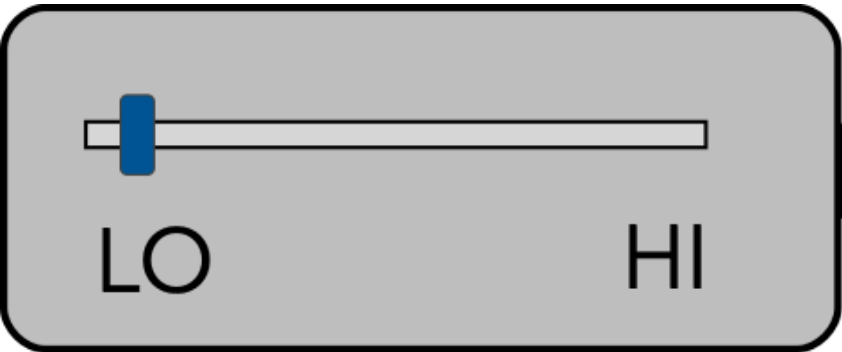


ultrashort
pulses

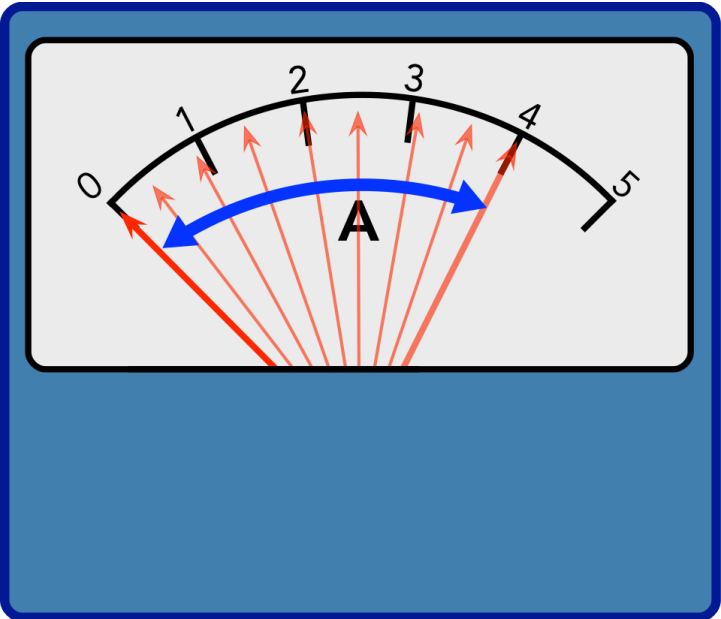
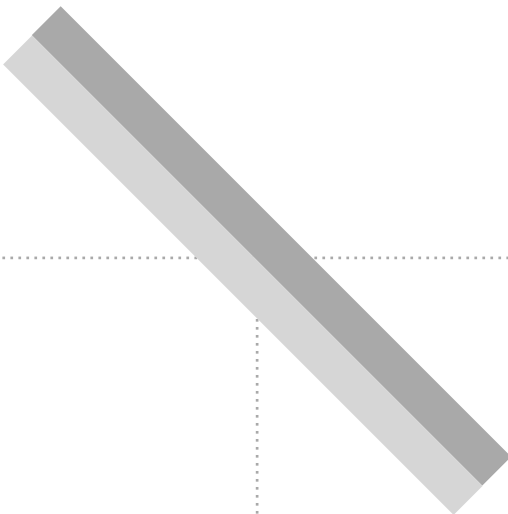
Experiment 2



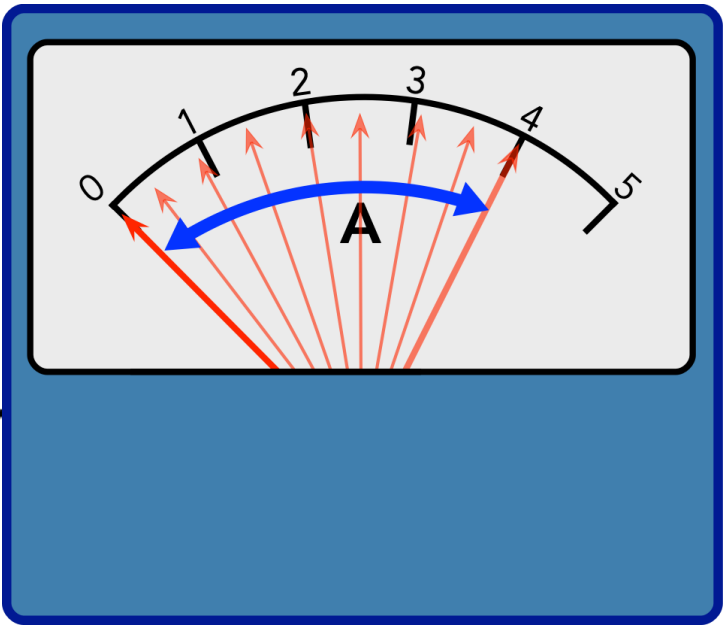
Experiment 2



Low intensity

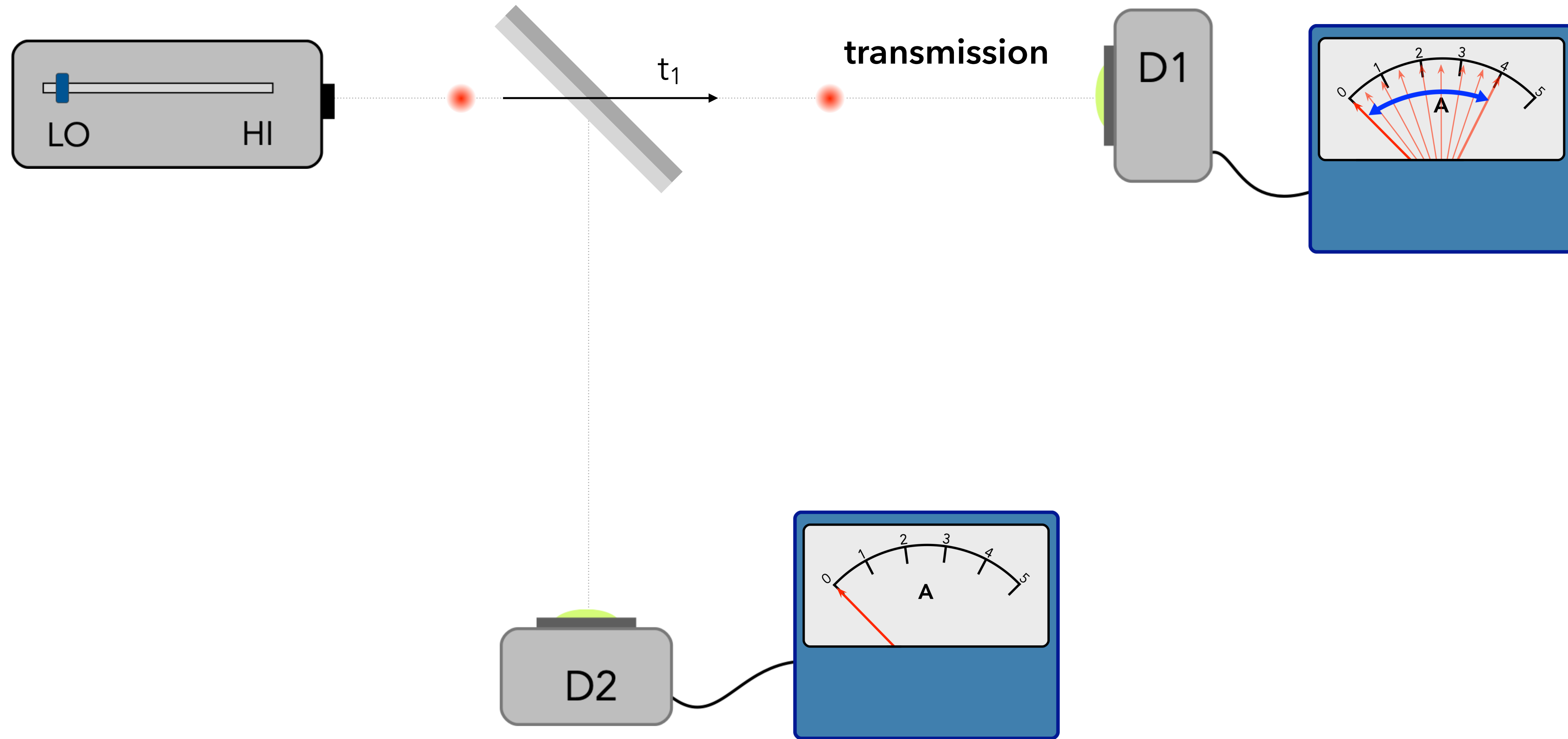


ultrashort
pulses

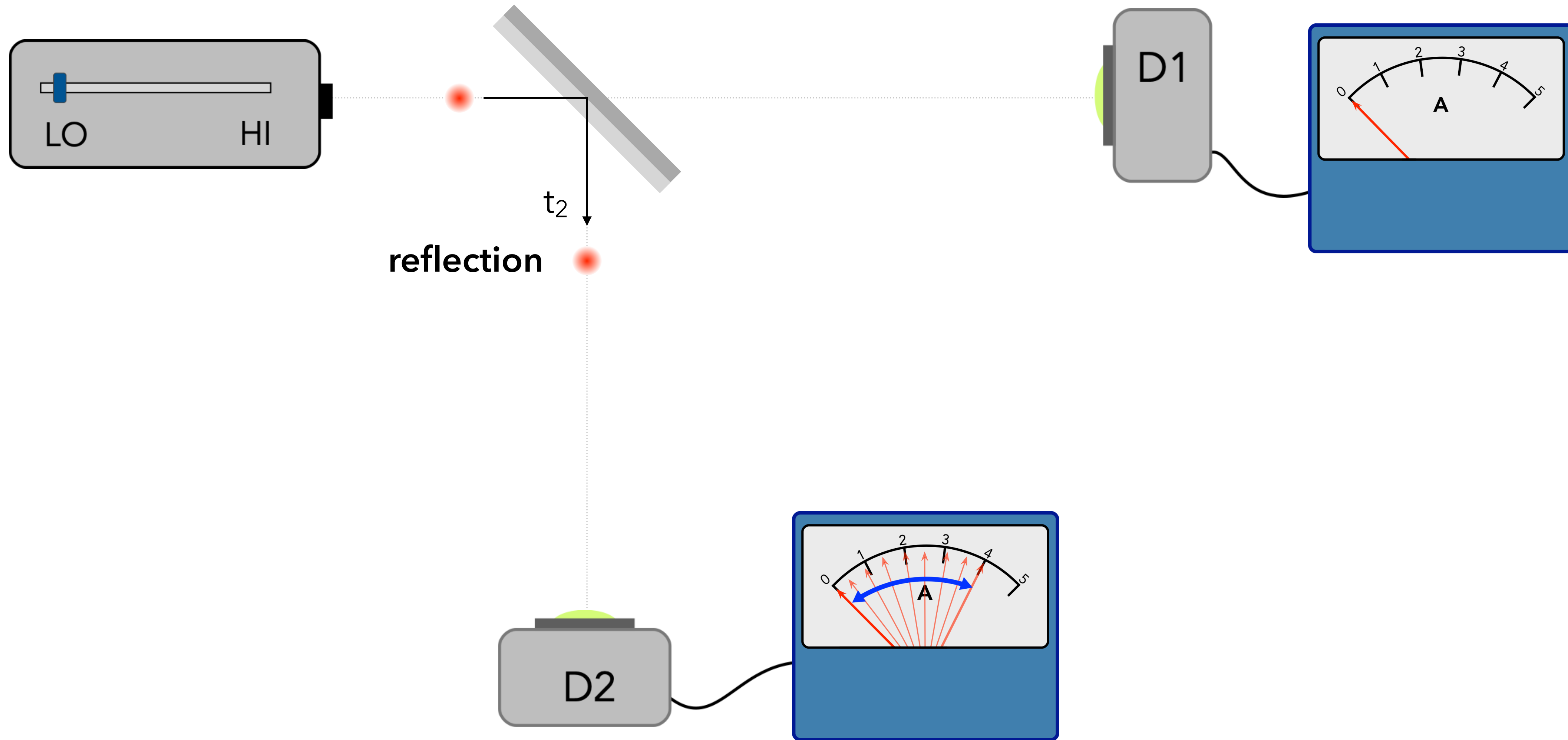


ultrashort
pulses

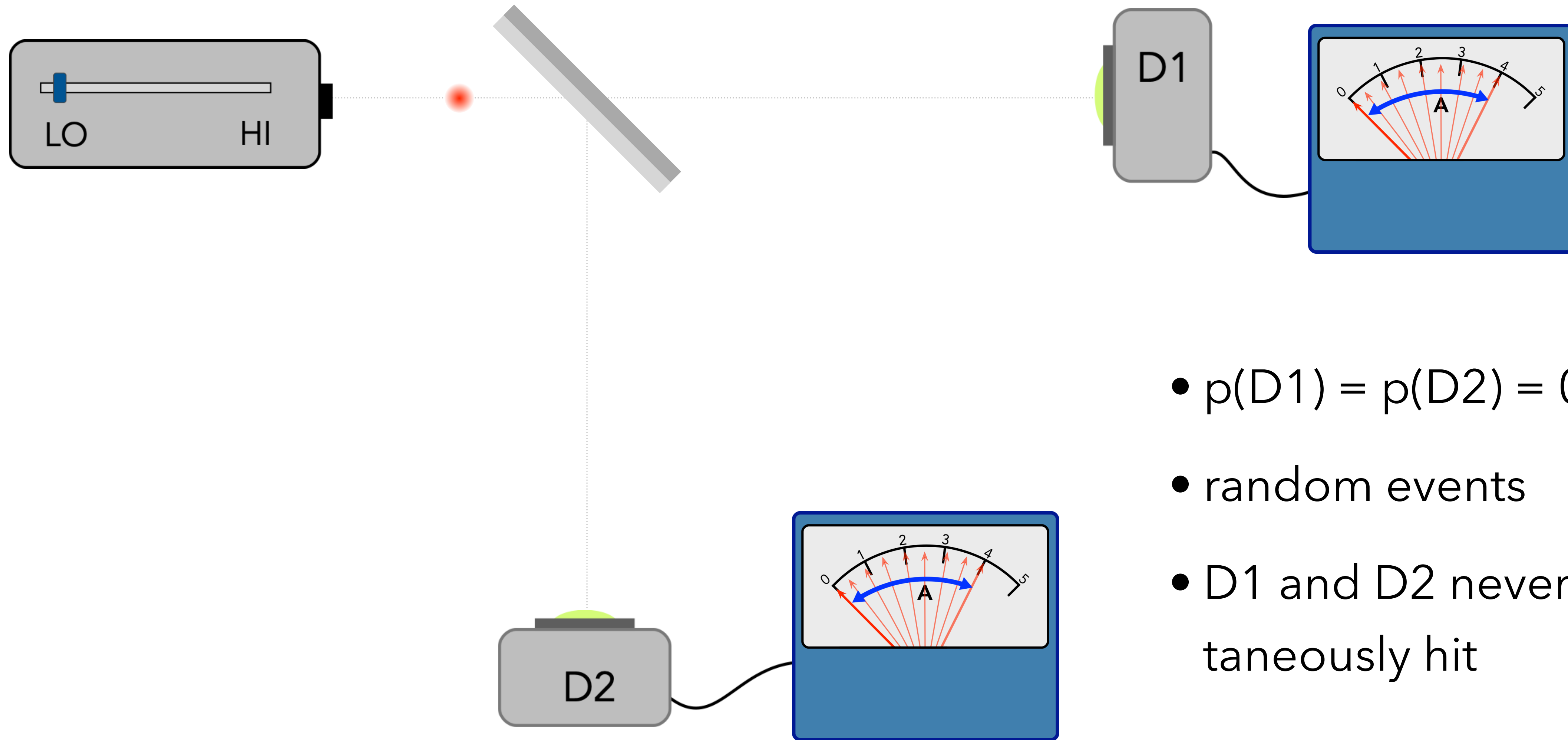
Experiment 2



Experiment 2



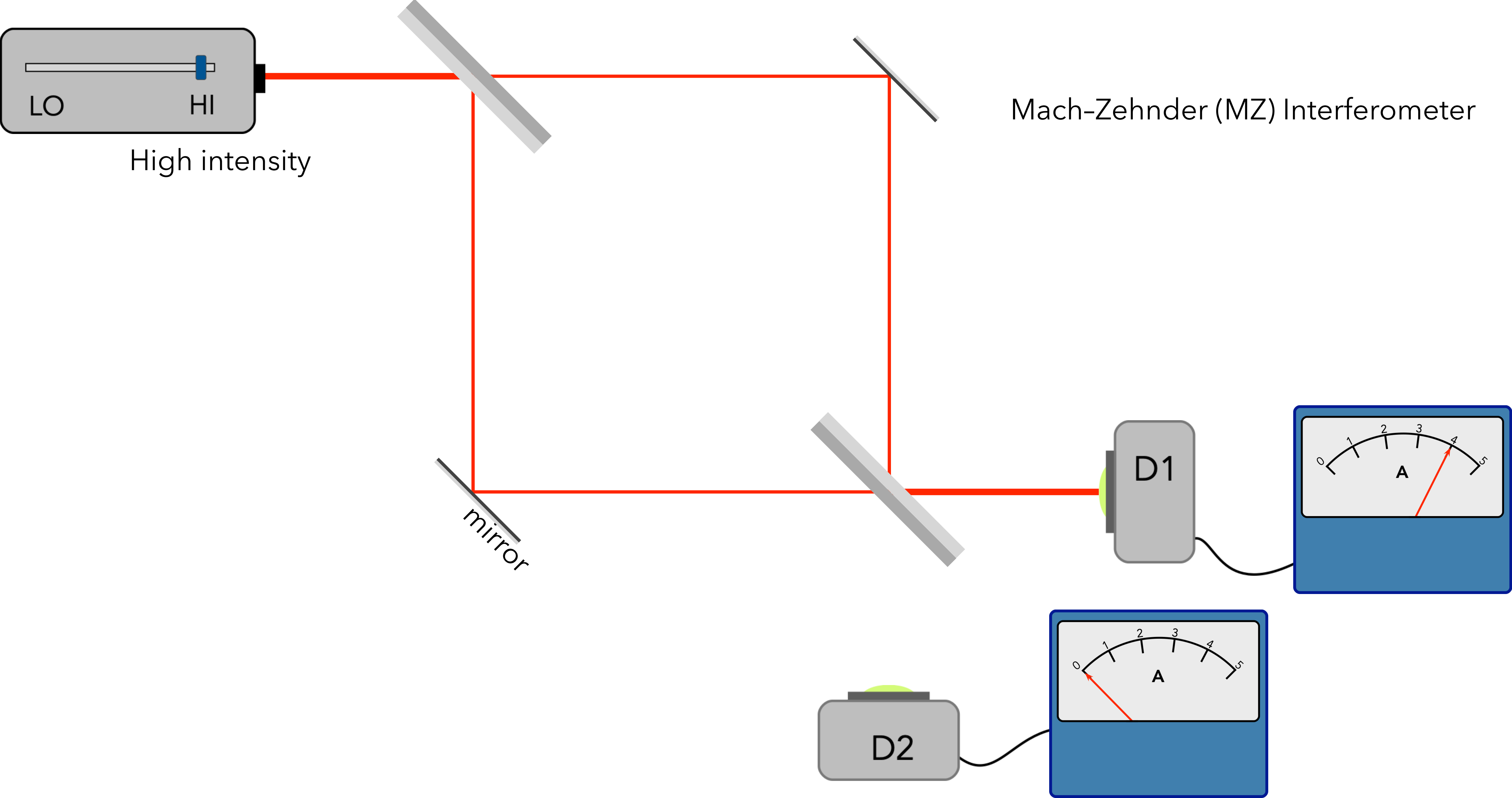
Experiment 2



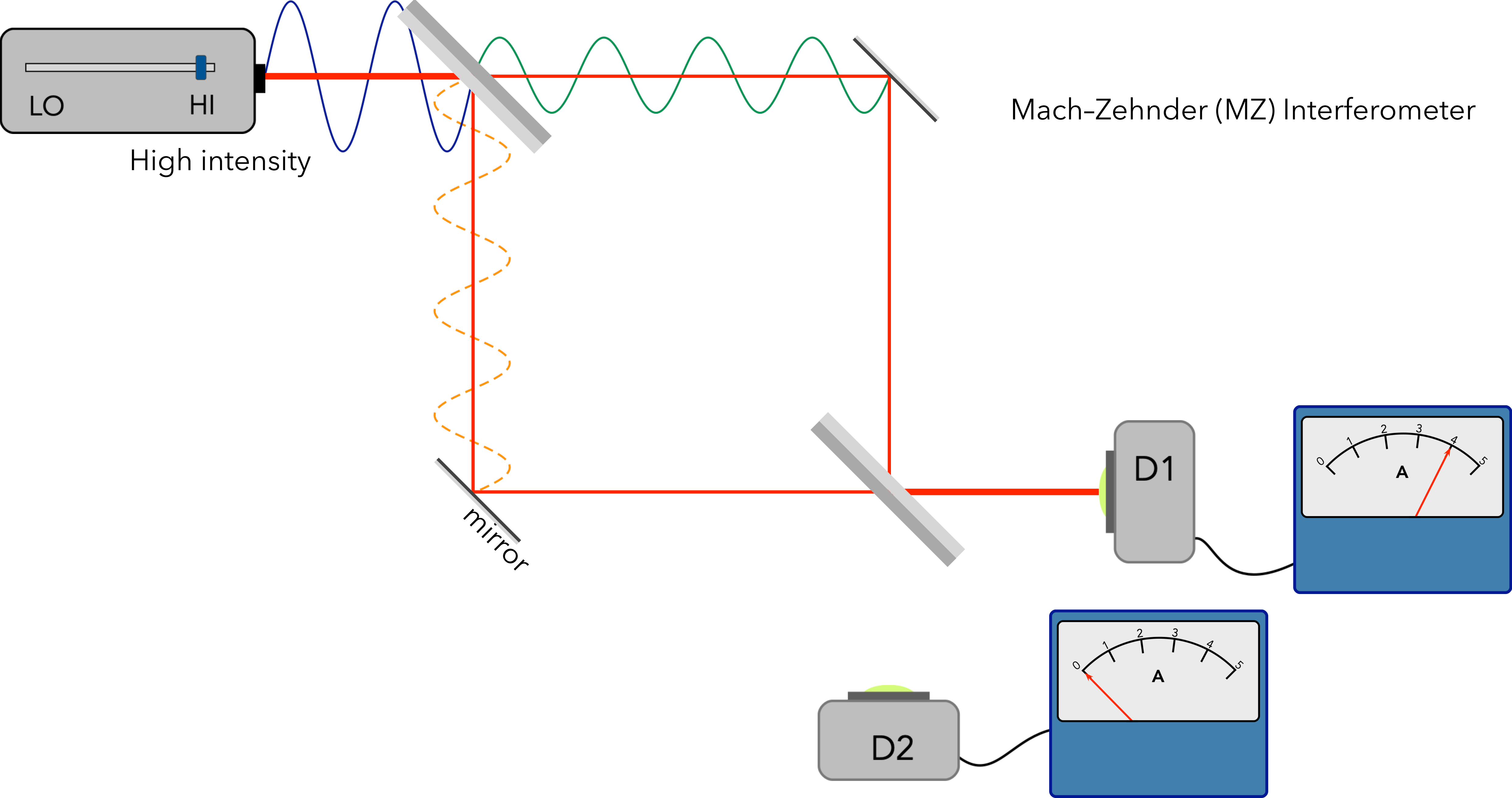
- $p(D1) = p(D2) = 0.5$
- random events
- D1 and D2 never simultaneously hit

light packets or "photons"

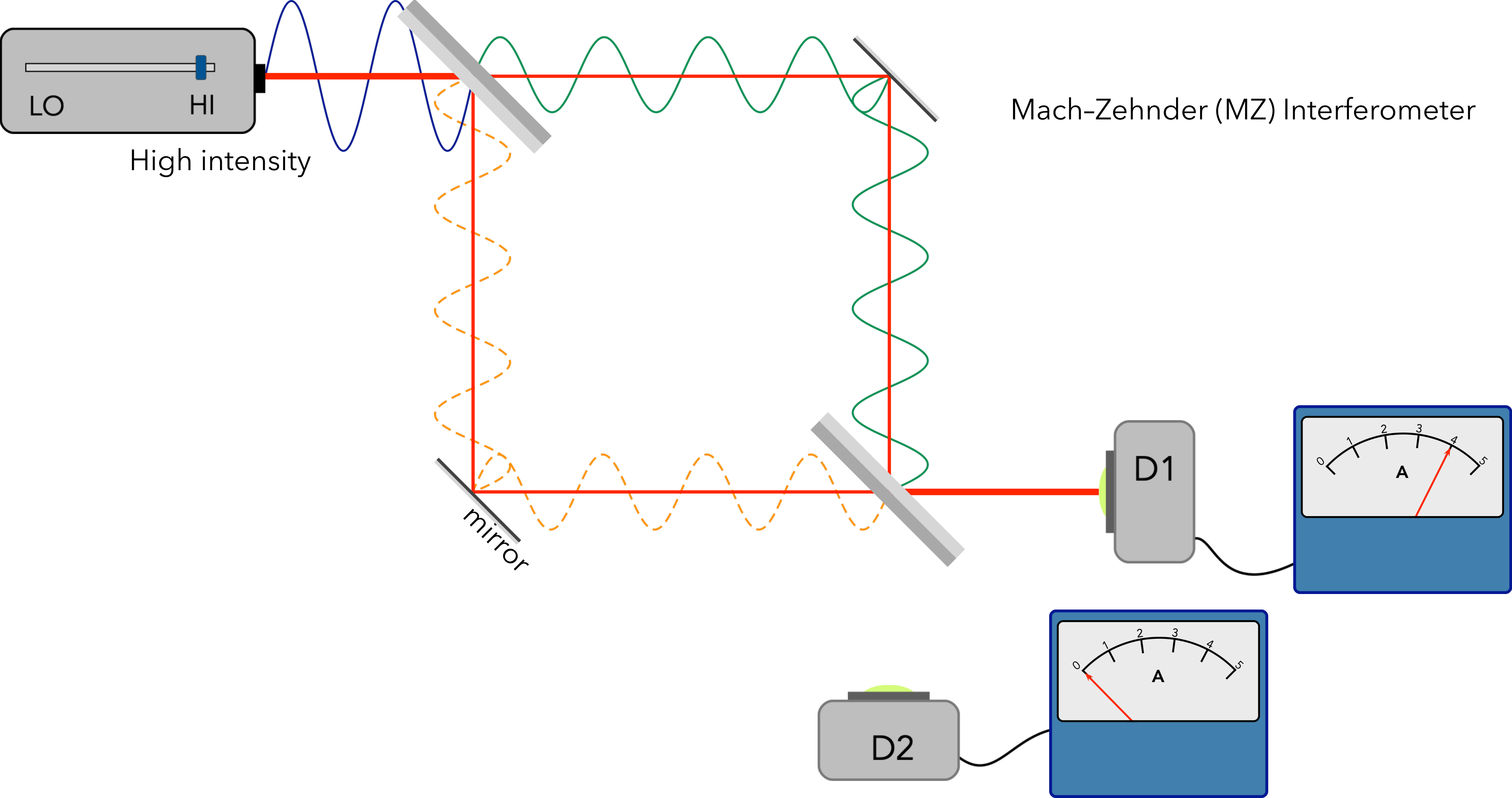
Experiment 3



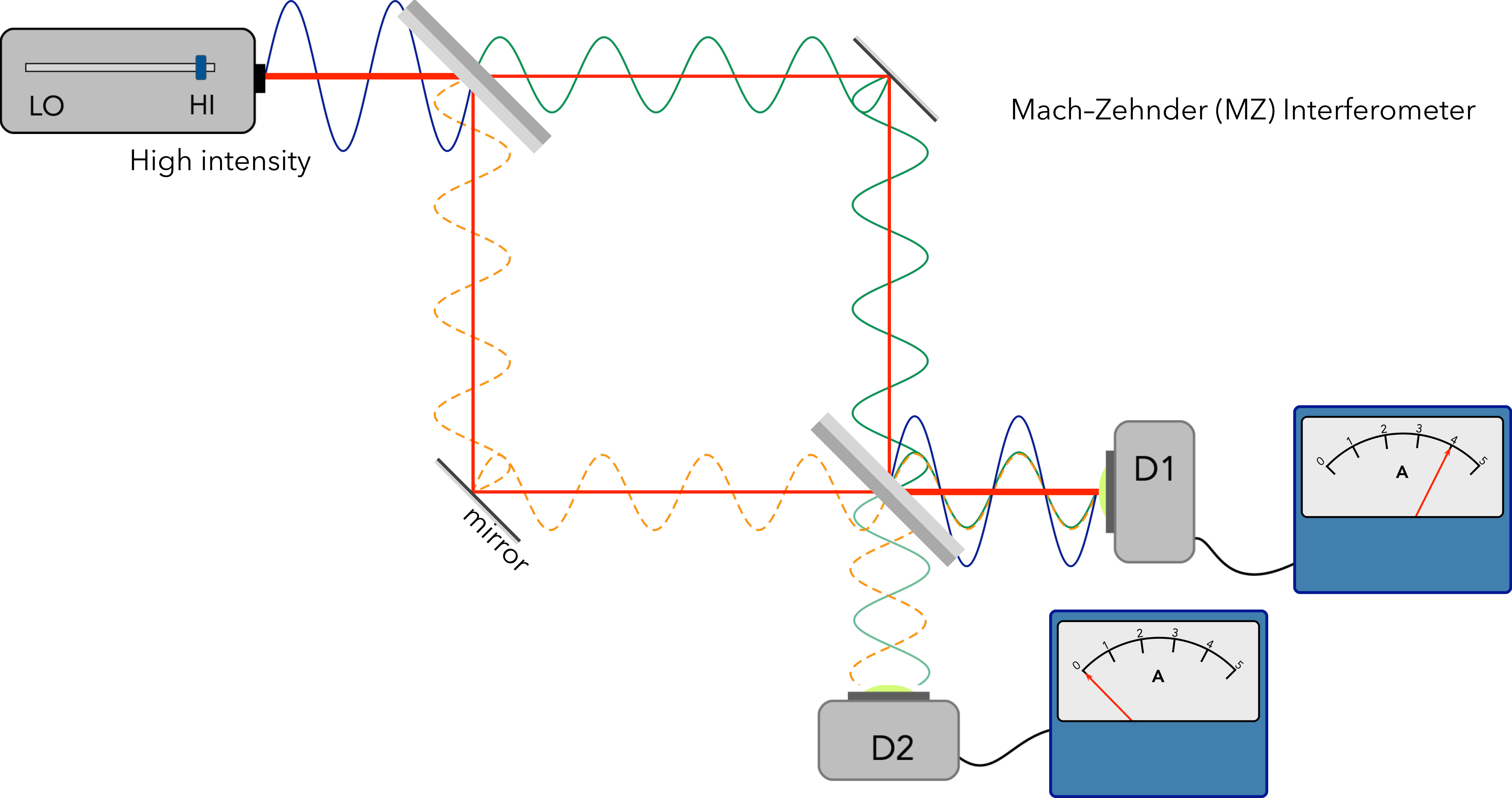
Experiment 3



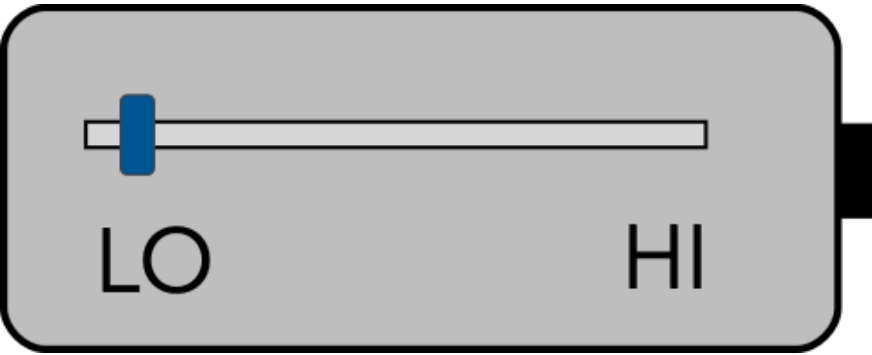
Experiment 3



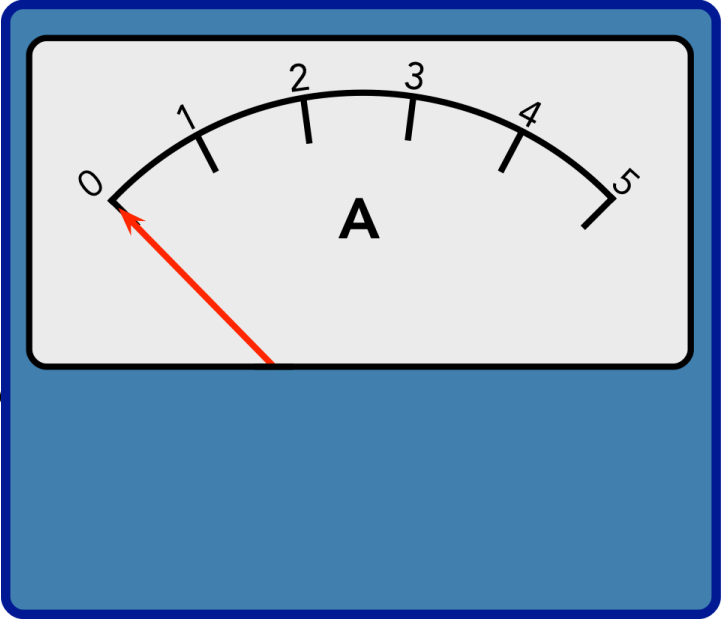
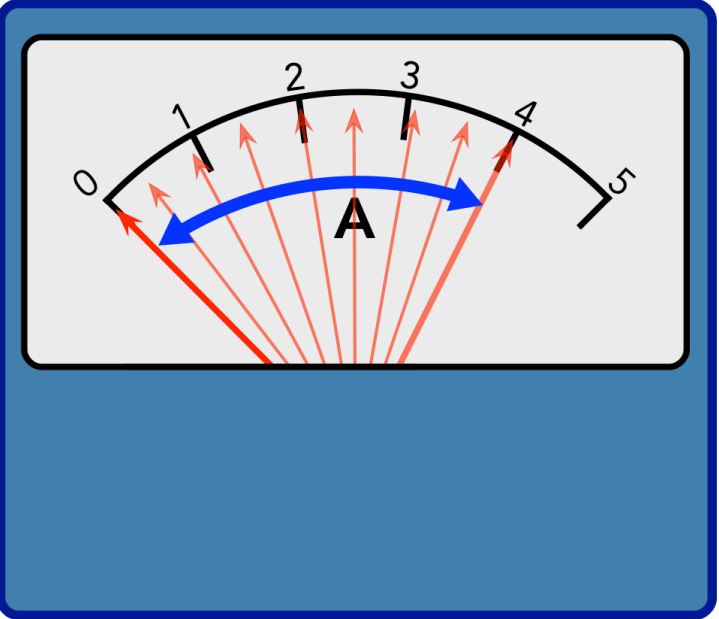
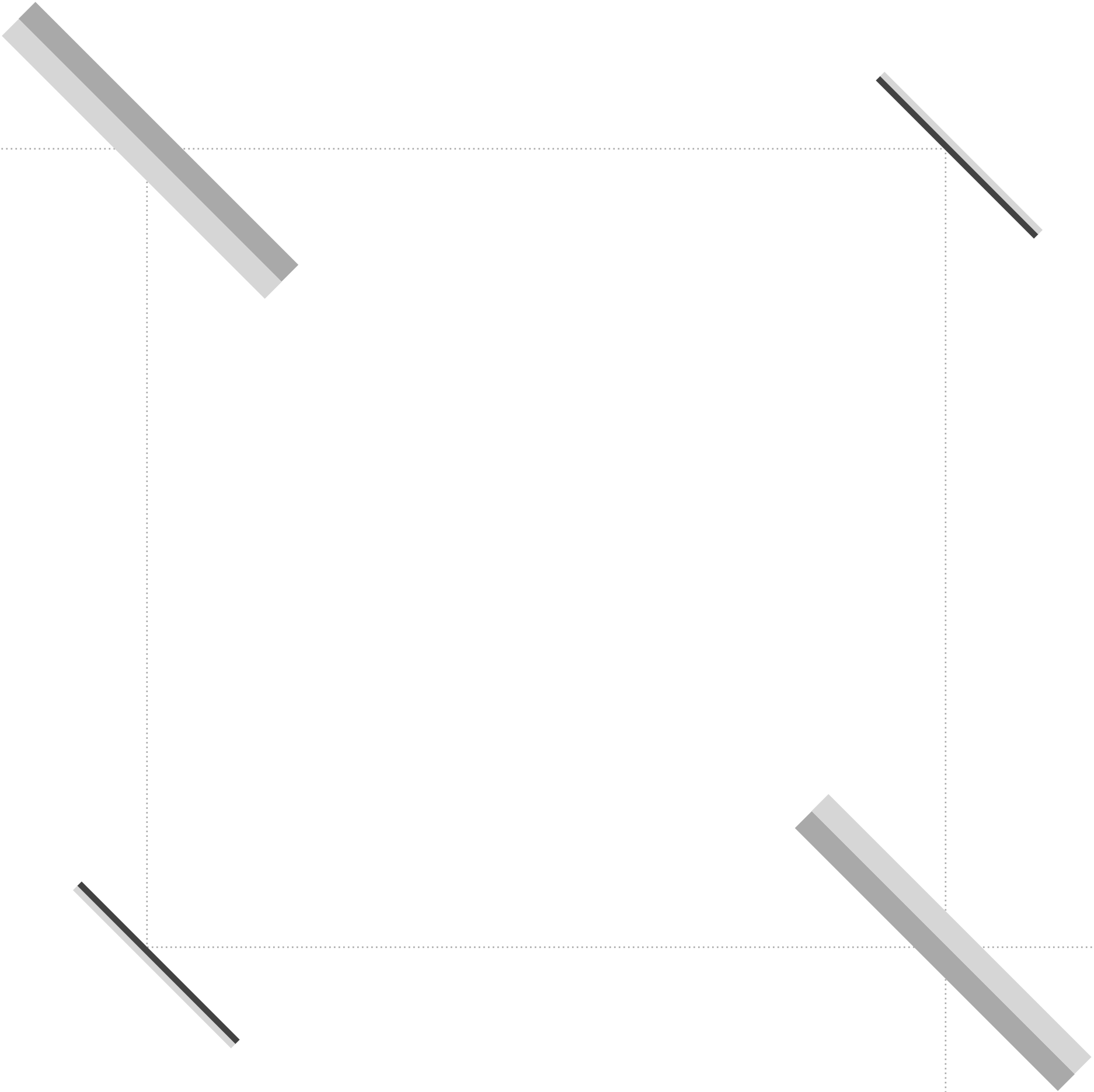
Experiment 3



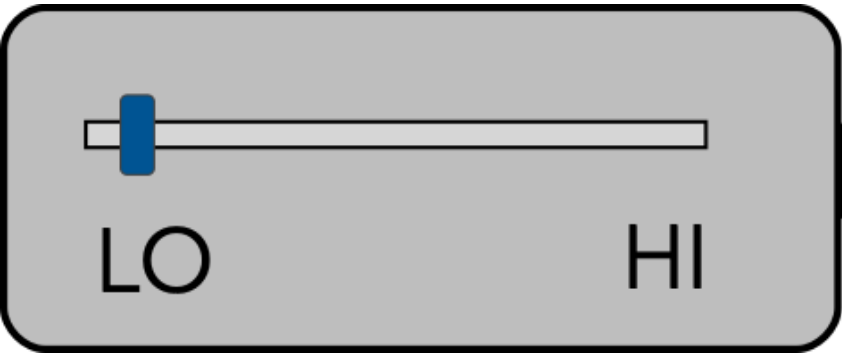
Experiment 3



Low intensity

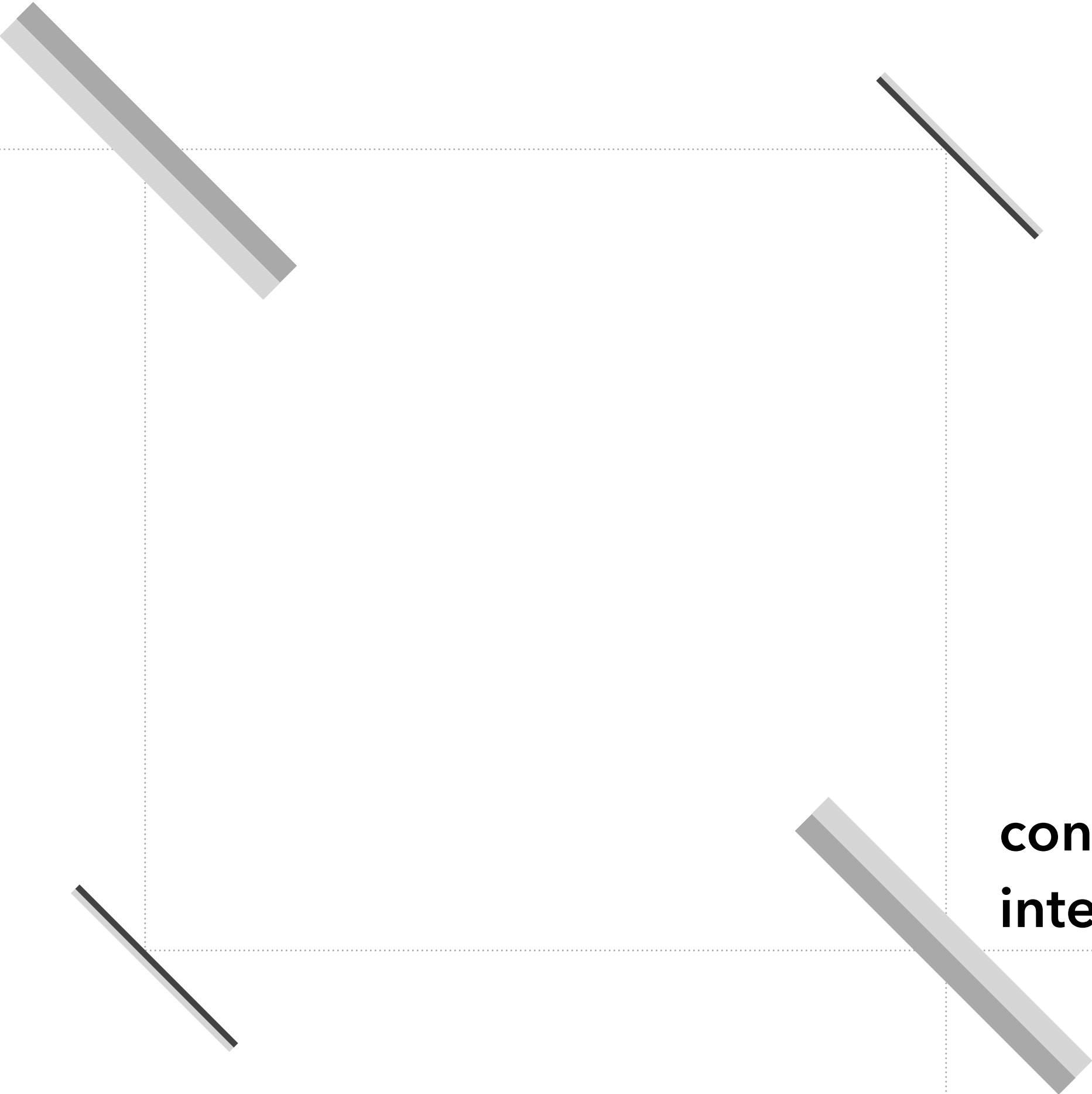


Experiment 3



Low intensity

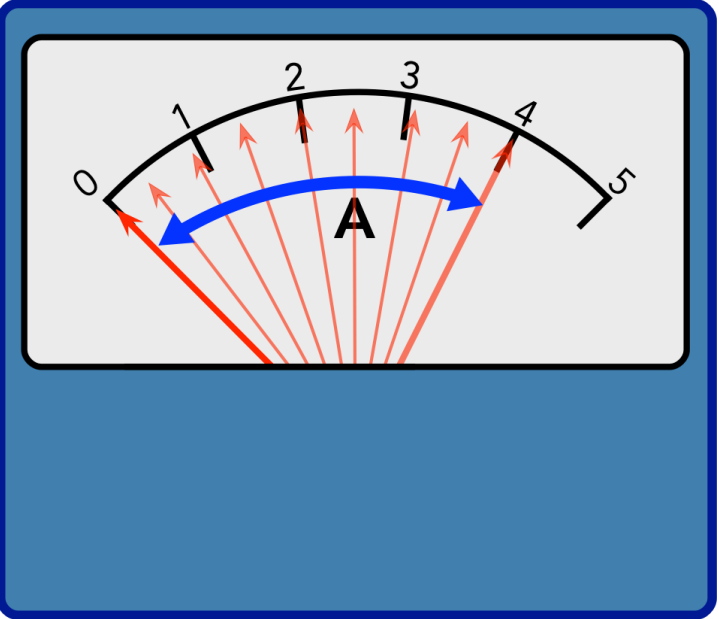
waves ?



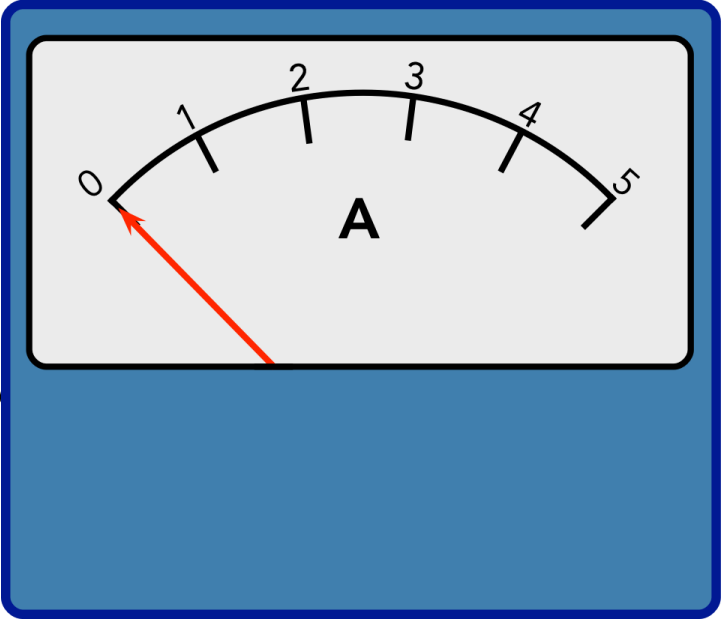
constructive interference



short pulses

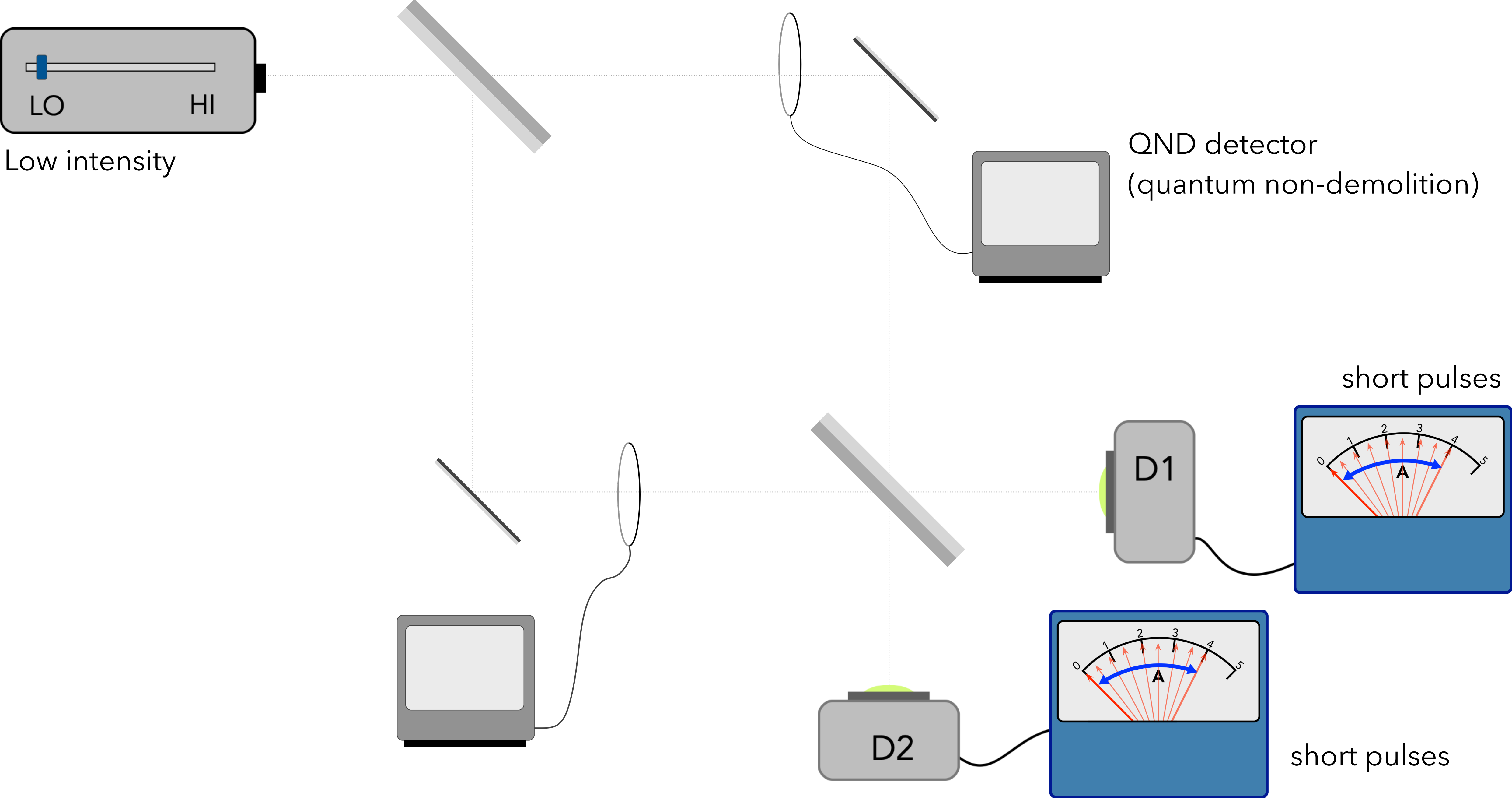


destructive interference



silent

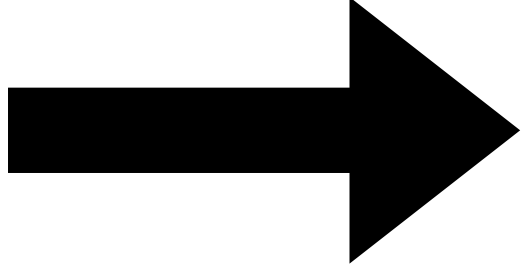
Experiment 3a



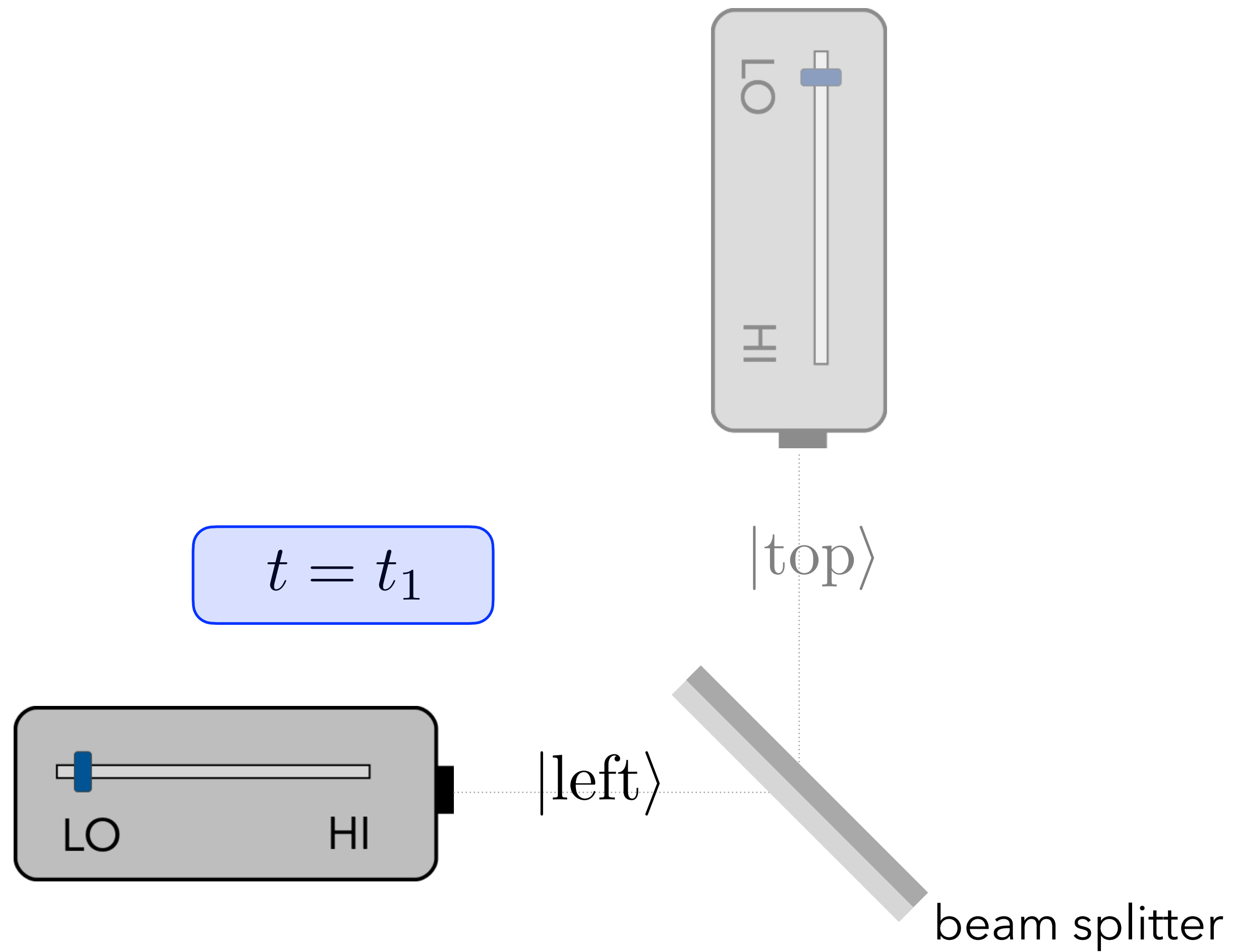
Photons

- originate pulses in the photodetectors
- indivisible light packets
- single photons can cause interference
- particle / wave duality
- measuring changes the system

Photons

- originate pulses in the photodetectors
- indivisible light packets
- single photons can cause interference
- ~~particle~~ / ~~wave~~ duality  new manifestation of matter: **quantum**
- measuring changes the system

Quanta of light: A mathematical description

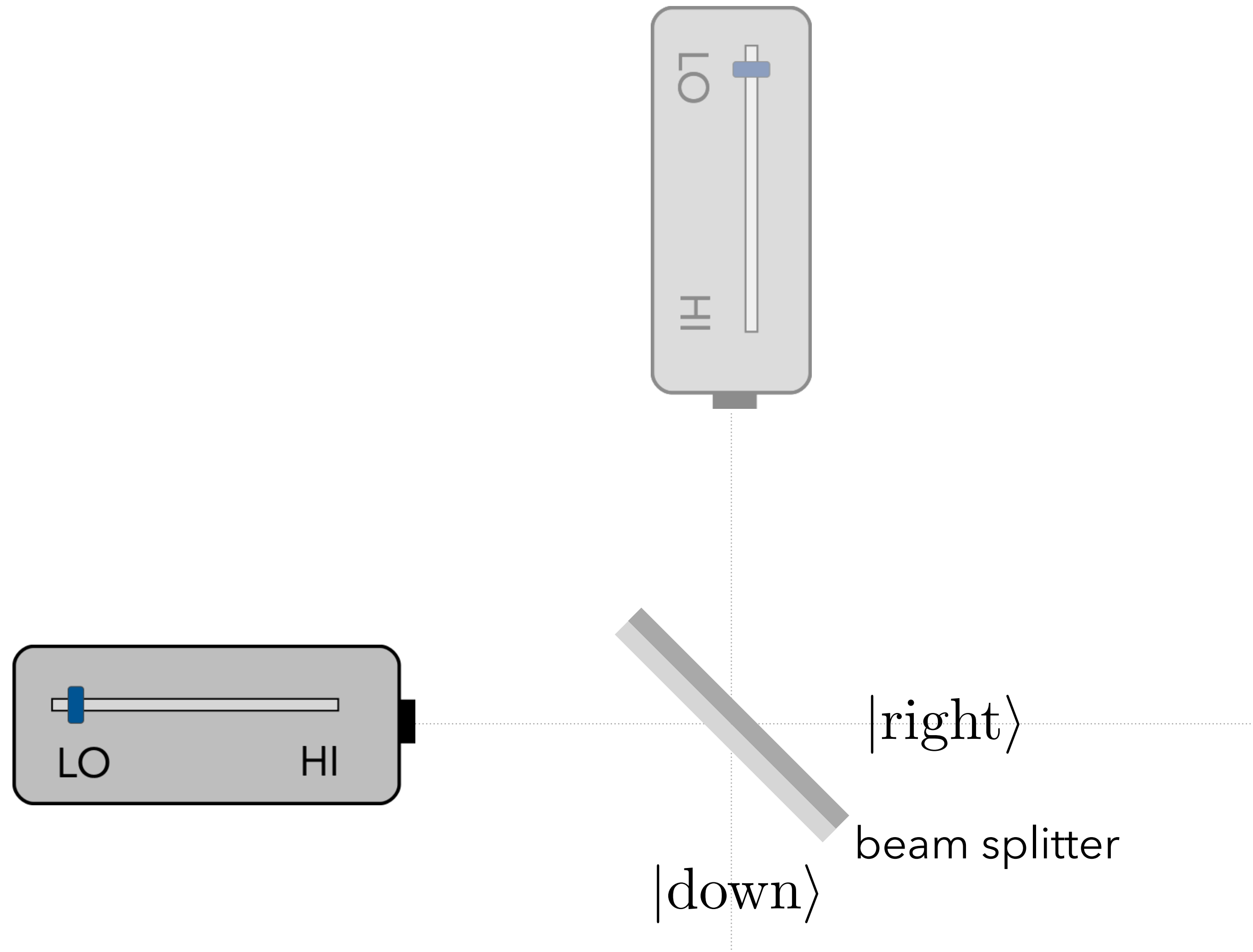


$$|\text{photon}\rangle_{t_1} = |\text{left}\rangle$$

Dirac notation

$|\dots\rangle$ ket

Quanta of light: A mathematical description



Dirac notation

$|\dots\rangle$ ket

Quanta of light: A mathematical description

$$t_1 : |\text{photon}\rangle = |\text{left}\rangle$$

$$t_2 : |\text{photon}\rangle = a|\text{right}\rangle + b|\text{down}\rangle$$

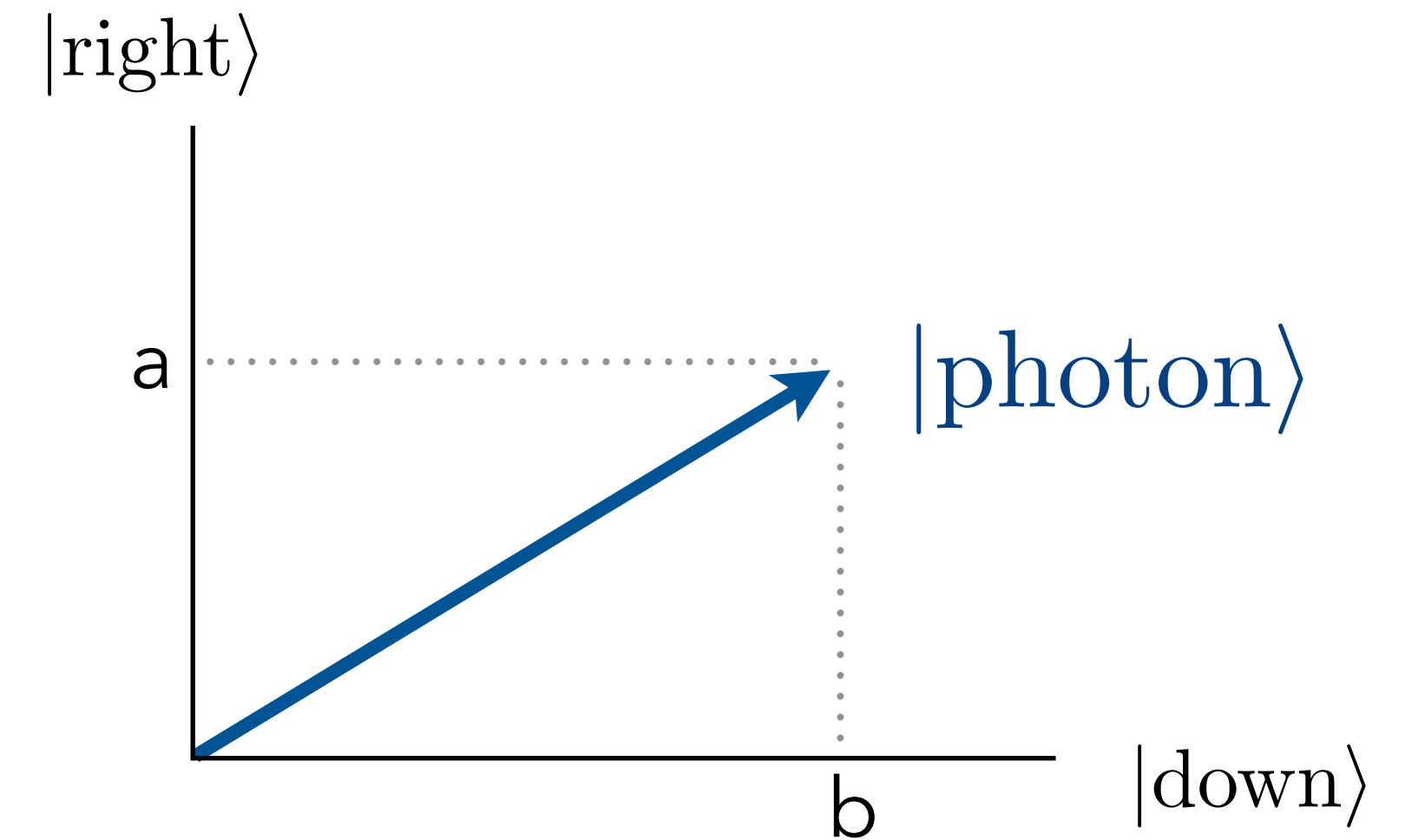
Quanta of light: A mathematical description

$$t_1 : |\text{photon}\rangle = |\text{left}\rangle$$

$$t_2 : |\text{photon}\rangle = a|\text{right}\rangle + b|\text{down}\rangle$$

$$|\text{right}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\text{down}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\text{photon}\rangle_{t_2} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$



representation of the photon in an
abstract vector space of quantum states

- 2 dimensional
- basis: $\{|\text{right}\rangle, |\text{down}\rangle\}$

Quanta of light: A mathematical description

$$t_2 : |\text{photon}\rangle = a|\text{right}\rangle + b|\text{down}\rangle$$

Probabilistic interpretation of the quantum state:

(Max Born, 1926)

$$p(D1) = |a|^2$$

$$p(D2) = |b|^2$$

$$p(D_1 + D_2) = 1$$

$$|a|^2 + |b|^2 = 1$$

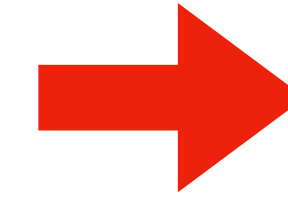
$$\Rightarrow a = b = \pm \frac{1}{\sqrt{2}}$$

$$p(D1) = p(D2) = 0.5$$

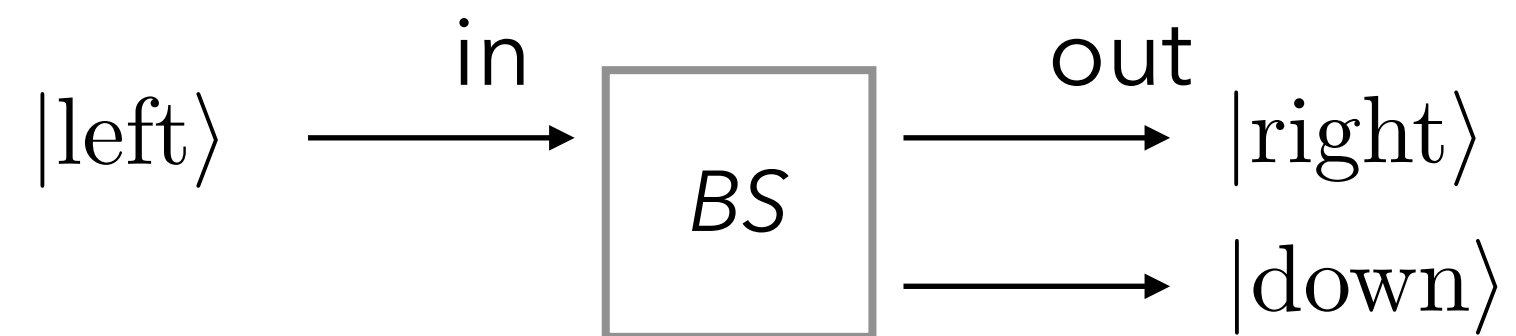
$$|\text{photon}\rangle_{t_2} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \pm \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

What about the *beam splitter*...

BS converts $\{|left\rangle, |top\rangle\}$ into $\{|right\rangle, |down\rangle\}$



$$U_{BS}|\text{in}\rangle_{t_1} = |\text{out}\rangle_{t_2}$$



$$U_{BS}|left\rangle = \frac{1}{\sqrt{2}}|right\rangle + \frac{1}{\sqrt{2}}|down\rangle$$

$$U_{BS} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

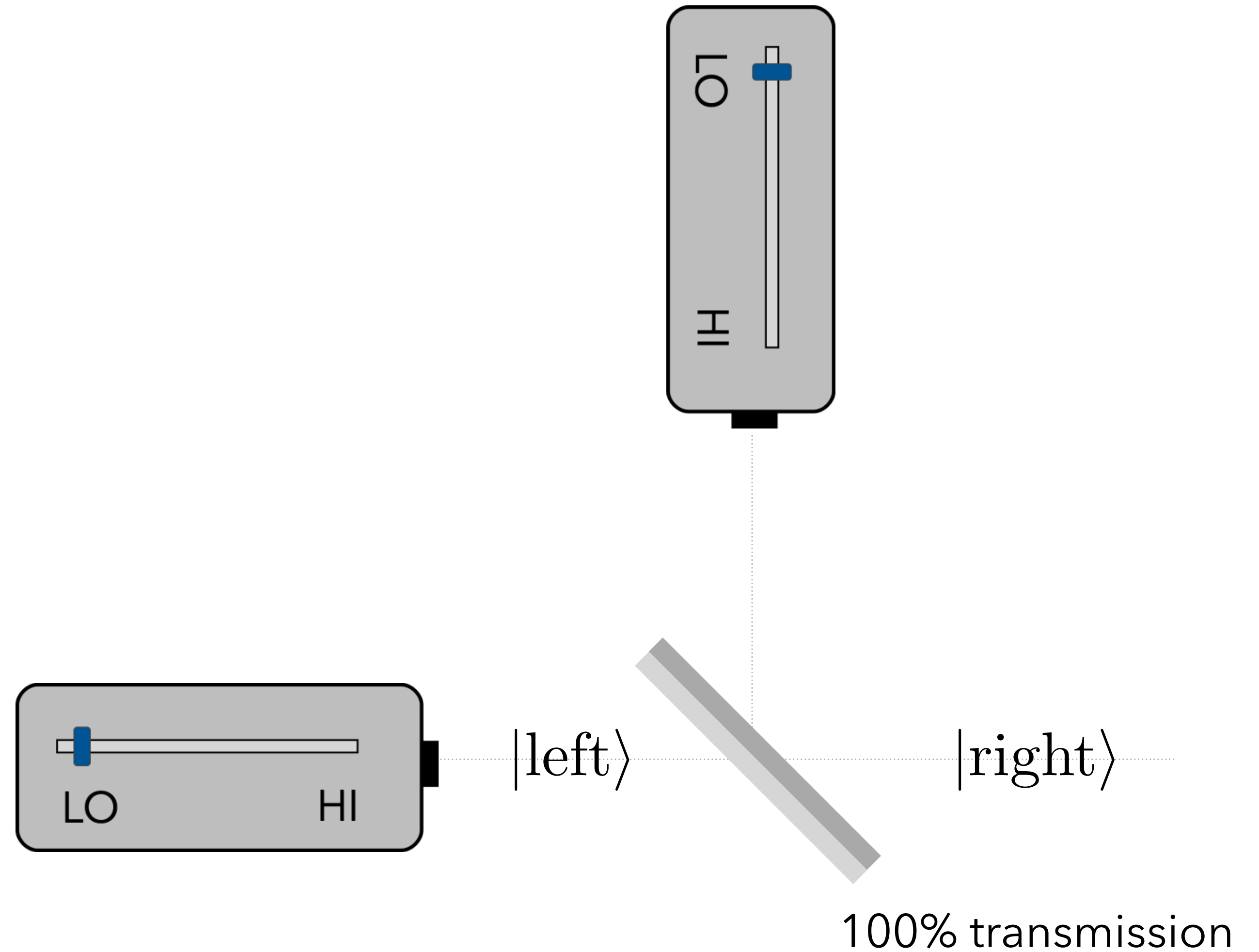
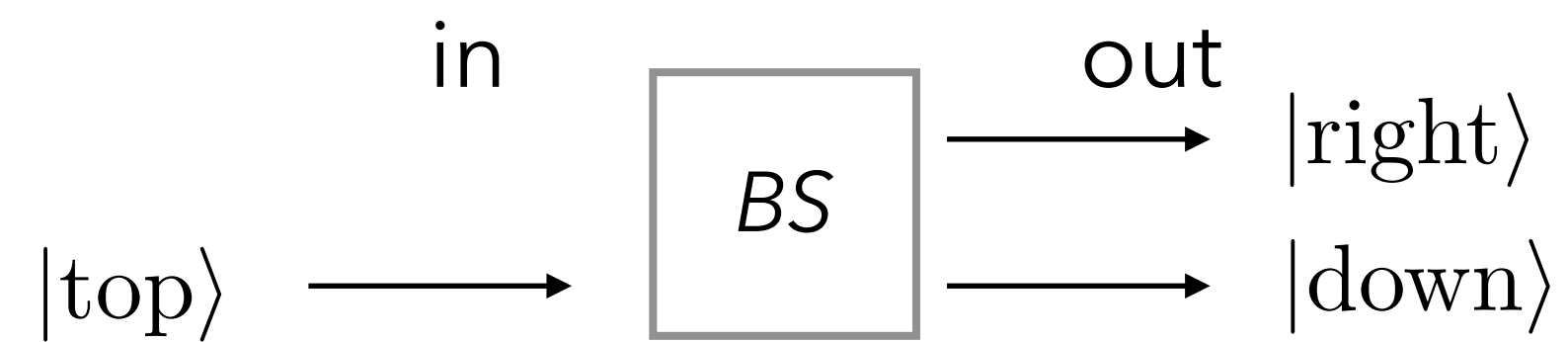
$$\text{with } U_{BS} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$$

$$\Rightarrow U_{BS} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} u_{11} \\ u_{21} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$u_{11} = u_{21} = \frac{1}{\sqrt{2}}$$

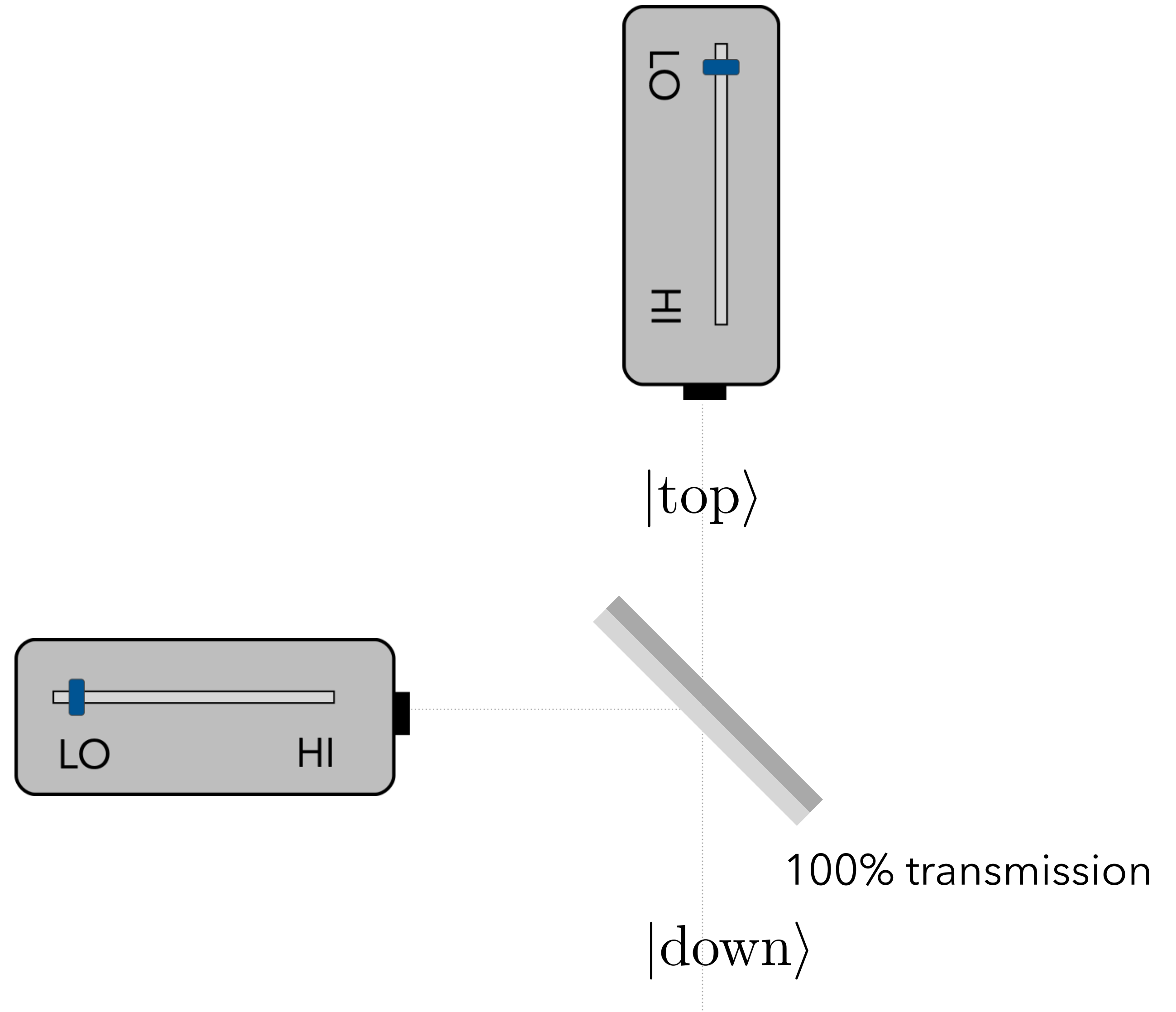
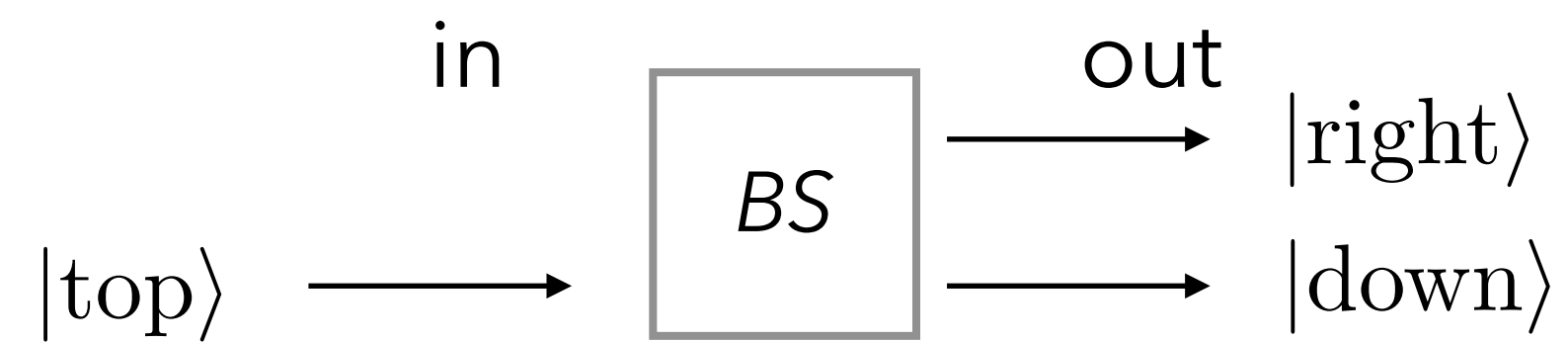
What about the *beam splitter*...

$$U_{BS} = \begin{pmatrix} \frac{1}{\sqrt{2}} & u_{12} \\ \frac{1}{\sqrt{2}} & u_{22} \end{pmatrix}$$



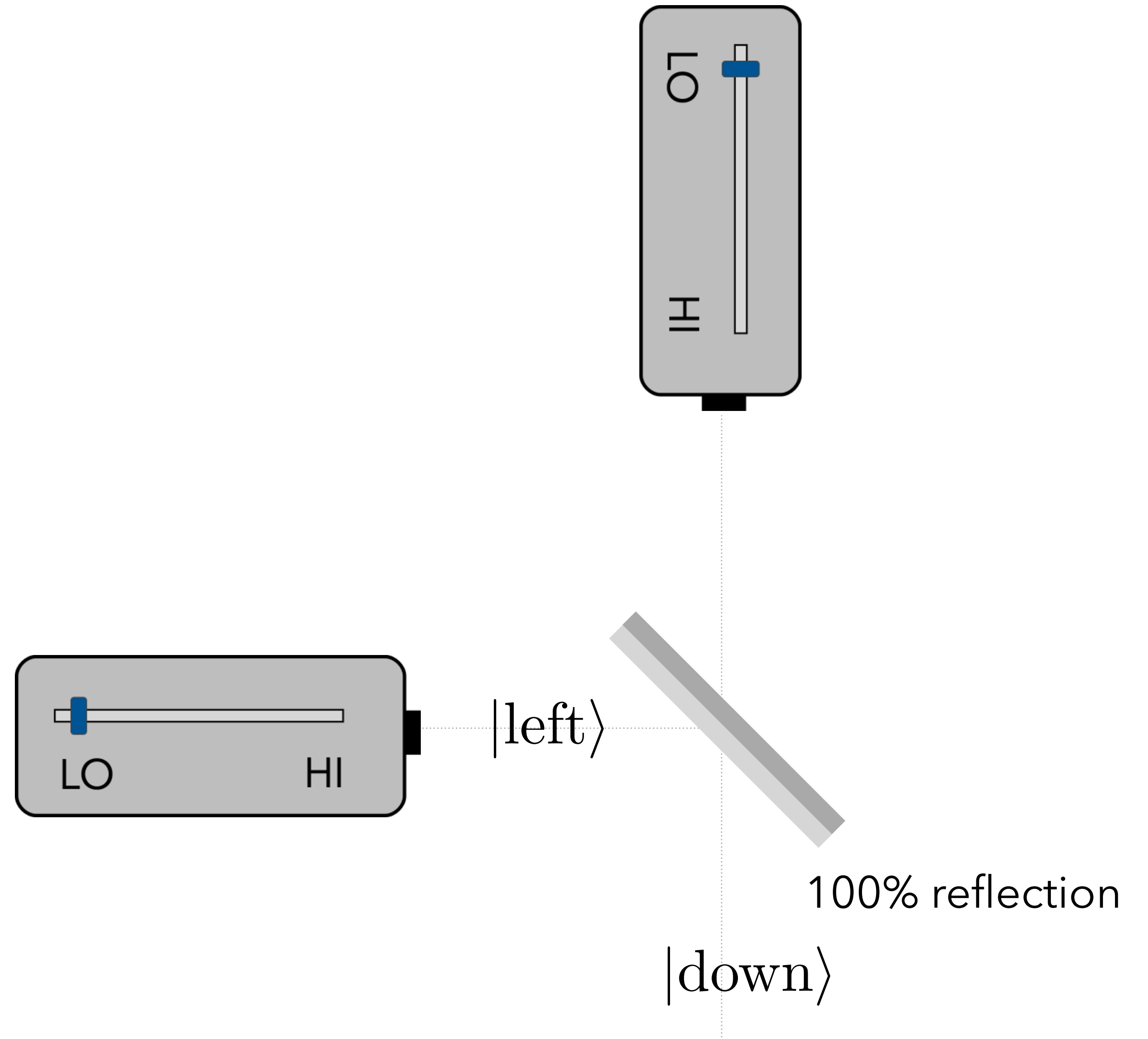
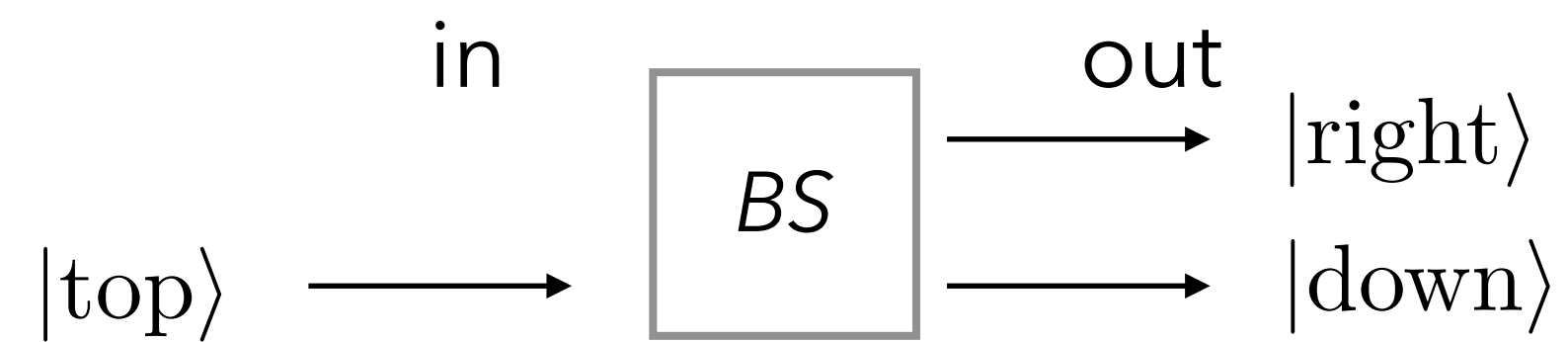
What about the *beam splitter*...

$$U_{BS} = \begin{pmatrix} \frac{1}{\sqrt{2}} & u_{12} \\ \frac{1}{\sqrt{2}} & u_{22} \end{pmatrix}$$



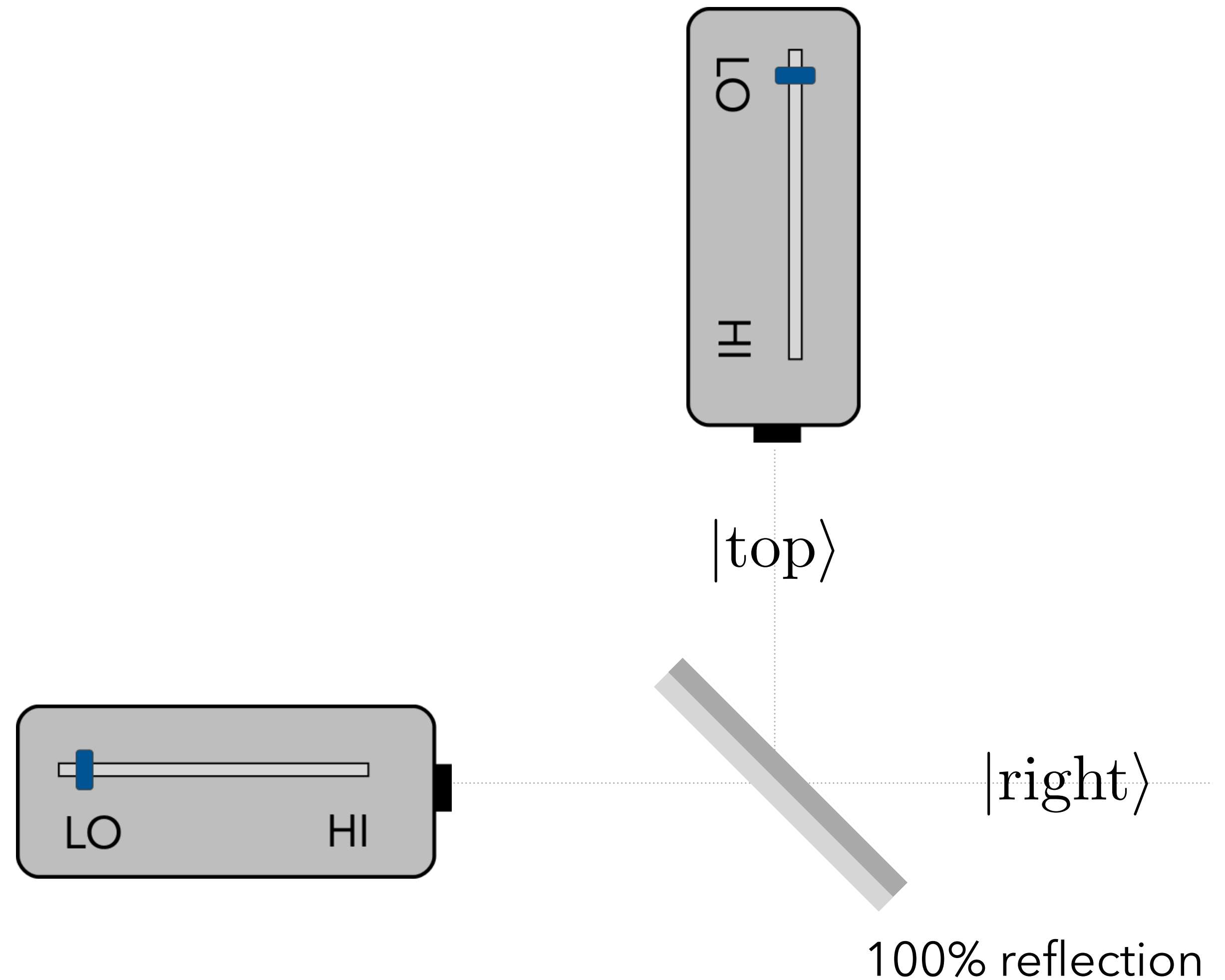
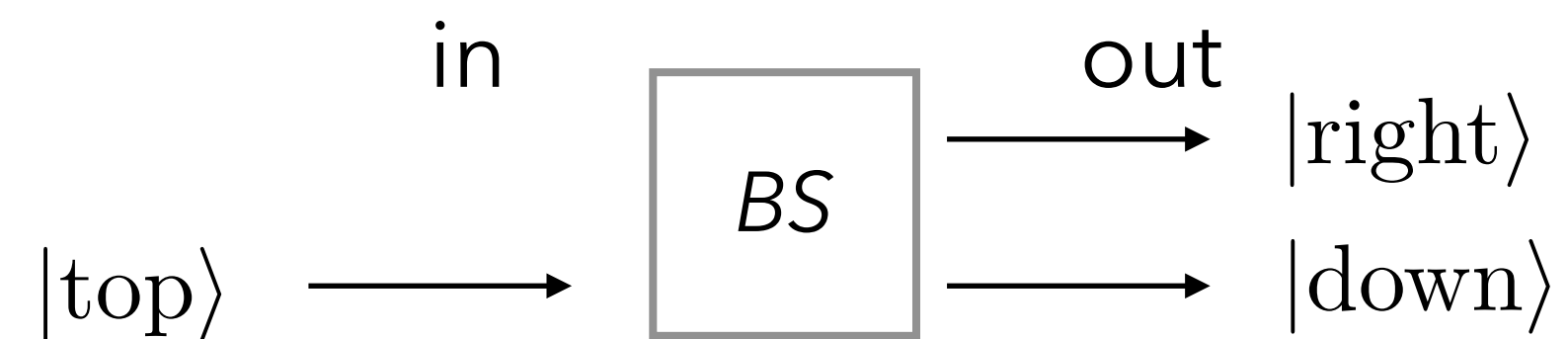
What about the *beam splitter*...

$$U_{BS} = \begin{pmatrix} \frac{1}{\sqrt{2}} & u_{12} \\ \frac{1}{\sqrt{2}} & u_{22} \end{pmatrix}$$



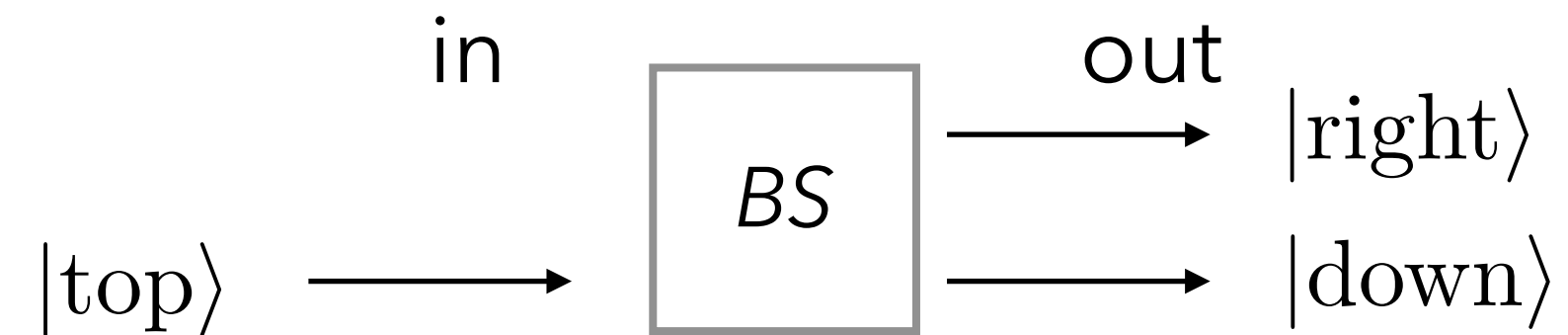
What about the *beam splitter*...

$$U_{BS} = \begin{pmatrix} \frac{1}{\sqrt{2}} & u_{12} \\ \frac{1}{\sqrt{2}} & u_{22} \end{pmatrix}$$



Orthogonal input states are converted in orthogonal output states

What about the *beam splitter*...



$$U_{BS}|\text{left}\rangle = \frac{1}{\sqrt{2}}|\text{right}\rangle + \frac{1}{\sqrt{2}}|\text{down}\rangle$$

$$U_{BS}|\text{top}\rangle = \frac{1}{\sqrt{2}}|\text{right}\rangle - \frac{1}{\sqrt{2}}|\text{down}\rangle$$

$$U_{BS} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

with $U_{BS} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$

$$\Rightarrow U_{BS} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} u_{12} \\ u_{22} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$u_{12} = \frac{1}{\sqrt{2}} \quad u_{22} = -\frac{1}{\sqrt{2}}$$

$$U_{BS} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$|\text{in}\rangle = a|\text{left}\rangle + b|\text{top}\rangle$$

$$|\text{out}\rangle = U_{BS}|\text{in}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ a-b \end{pmatrix}$$

Interference: Introducing the phase

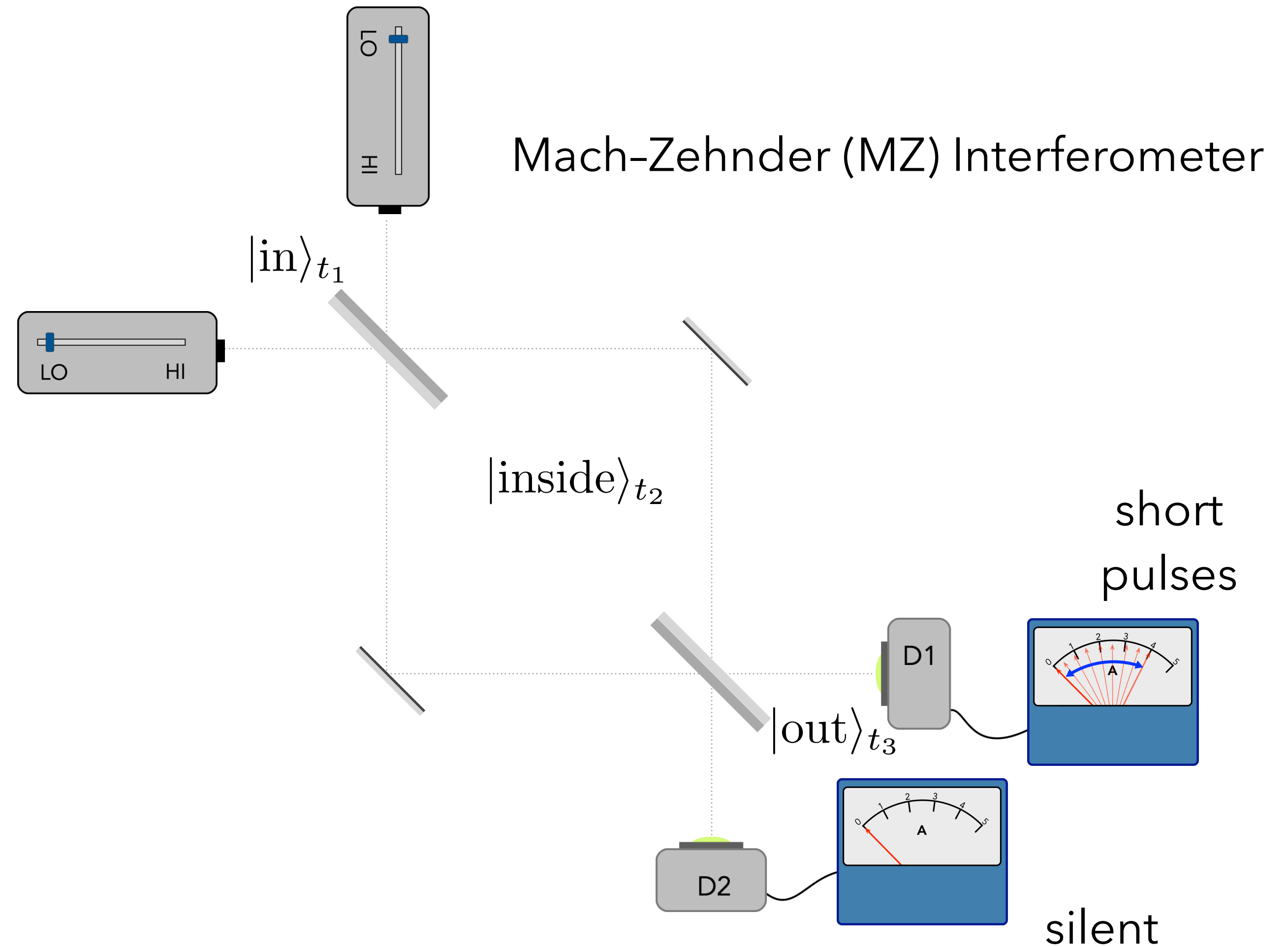
$$|\text{inside}\rangle_{t_2} = U_{BS}|\text{in}\rangle_{t_1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|\text{out}\rangle_{t_3} = U_{BS}|\text{inside}\rangle_{t_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1+1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\text{D1}\rangle_{t_3}$$

or more directly:

$$|\text{out}\rangle_{t_3} = U_{BS}U_{BS}|\text{in}\rangle_{t_1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\text{D1}\rangle_{t_3}$$



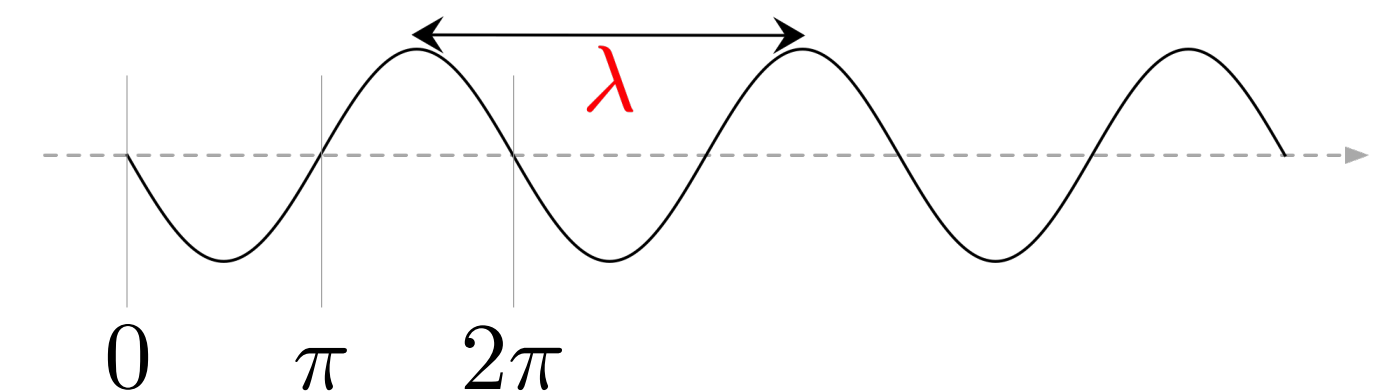
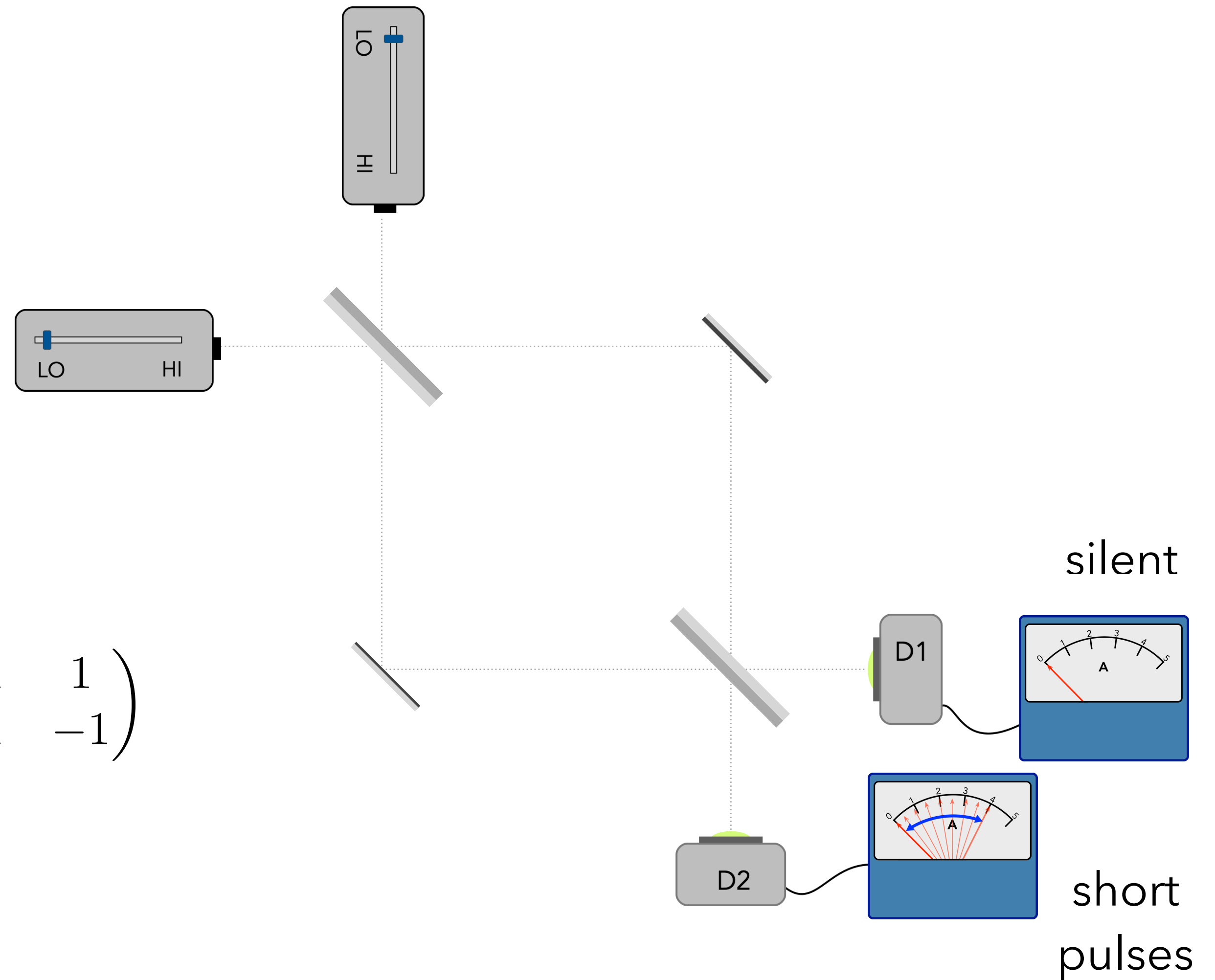
Interference: Introducing the phase

Increasing one pathway by $\lambda/2$

$$U_{\lambda/2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned} U_{BS}U_{\lambda/2}U_{BS} &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$$|\text{out}\rangle_{t_3} = U_{BS}U_{\lambda/2}U_{BS}|\text{in}\rangle_{t_1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\text{D2}\rangle_{t_3}$$



Interference: Introducing the phase

$$U_{\lambda/2}|\text{inside}\rangle_{t_2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

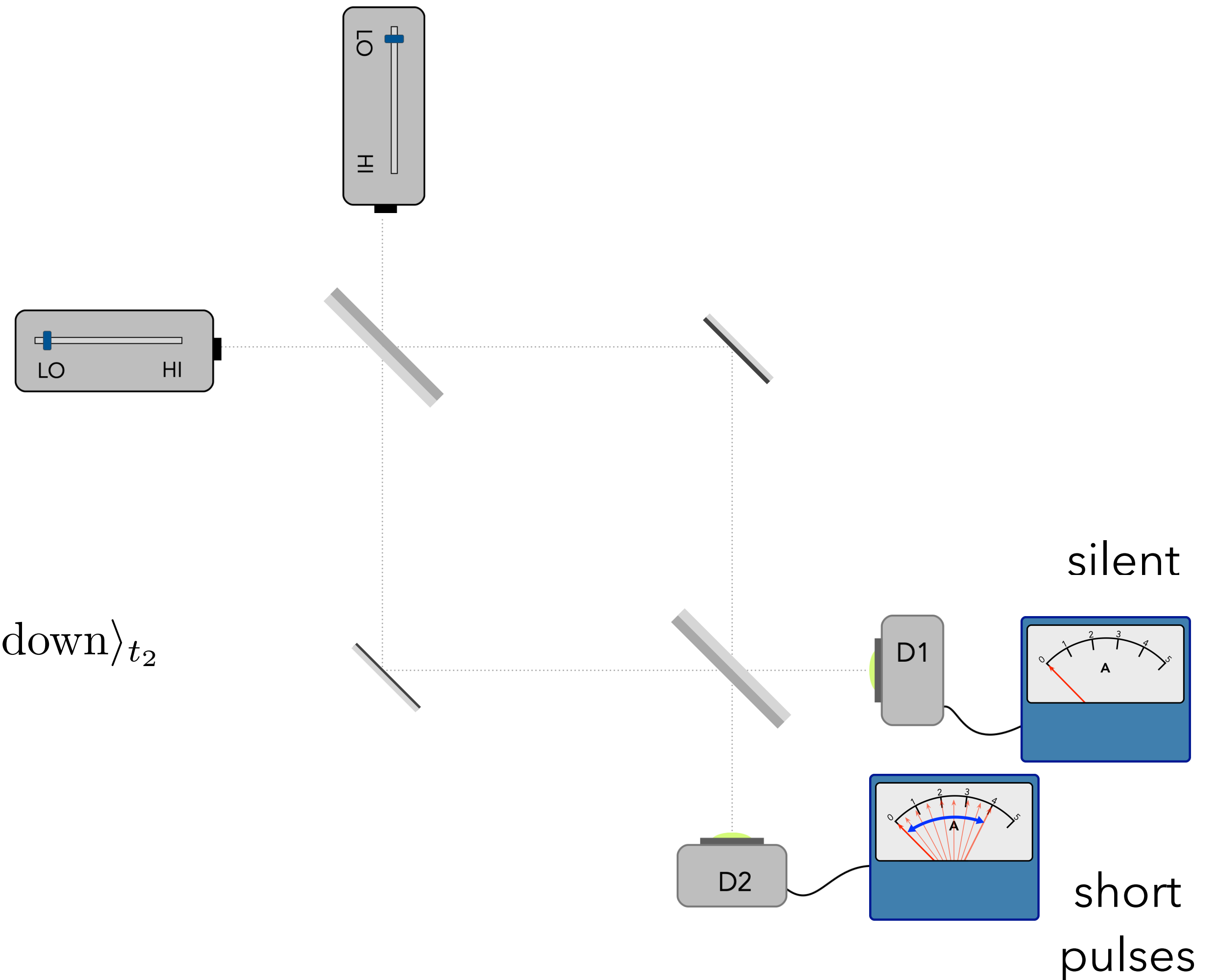
$$\frac{1}{\sqrt{2}}|\text{right}\rangle_{t_2} + \frac{1}{\sqrt{2}}|\text{down}\rangle_{t_2}$$

$$\frac{1}{\sqrt{2}}|\text{right}\rangle_{t_2} - \frac{1}{\sqrt{2}}|\text{down}\rangle_{t_2}$$

Generalizing: $U_\phi = \begin{pmatrix} 1 & 0 \\ 0 & \phi \end{pmatrix} \quad -1 \leq \phi \leq 1$

not norm-conserving

Better: $U_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \quad -\pi \leq \phi \leq \pi$



Interference: Introducing the phase

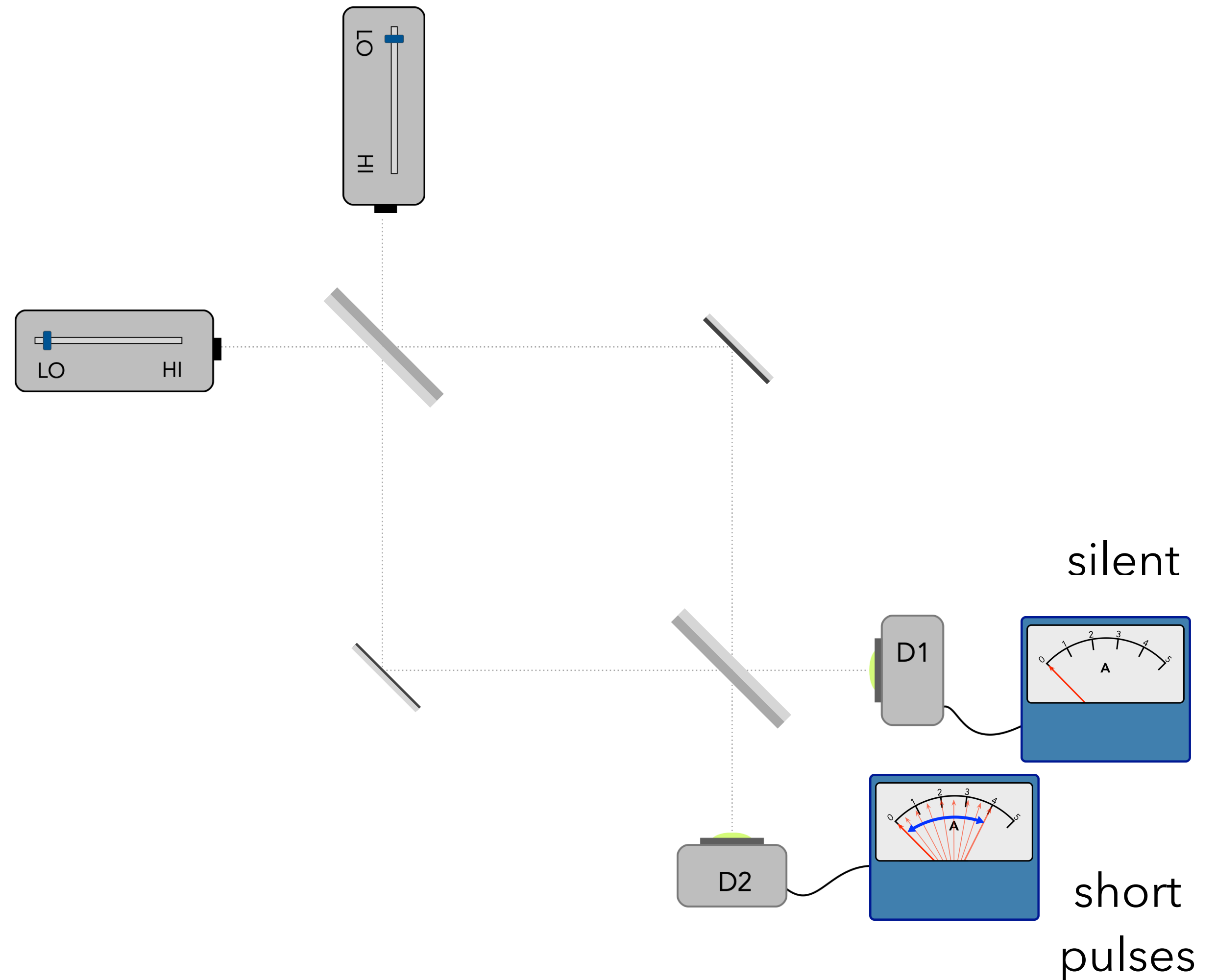
$$U_{BS}U_{\phi}U_{BS} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \times \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} & 1 - e^{i\phi} \\ 1 - e^{i\phi} & 1 + e^{i\phi} \end{pmatrix} = U_{MZ}$$

$$|\text{out}\rangle_{t_3} = U_{MZ}|\text{left}\rangle_{t_1} = \frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} \\ 1 - e^{i\phi} \end{pmatrix}$$

$$\phi = 0 \rightarrow |\text{out}\rangle_{t_3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\text{D1}\rangle_{t_3}$$

$$\phi = \pi \rightarrow |\text{out}\rangle_{t_3} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\text{D2}\rangle_{t_3}$$



Probabilities: How to calculate $p(D1)$ and $p(D2)$

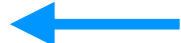
$$|\text{photon}\rangle_{t_2} = a|\text{right}\rangle + b|\text{down}\rangle$$

$$p(D1) = |a|^2 \quad ; \quad p(D2) = |b|^2$$

Dirac notation

$|\dots\rangle$ ket  vector

$\langle\dots|$ bra  complex conjugate

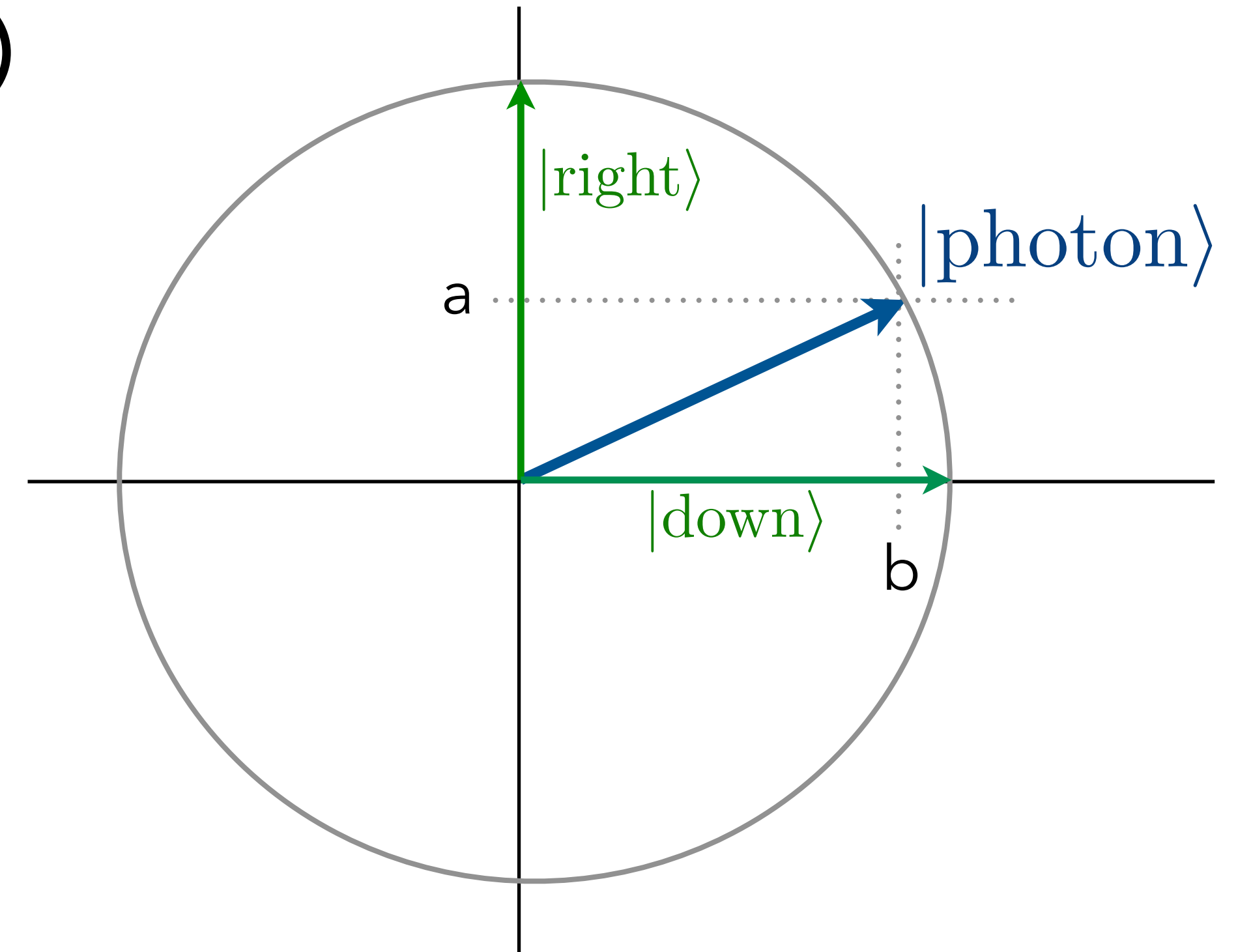
$\langle\dots|\dots\rangle$ bra-ket  scalar product

$$|\phi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\langle\phi| = (a^* \quad b^*)$$

$$|\psi\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$\langle\phi|\psi\rangle = (a^* \quad b^*) \begin{pmatrix} c \\ d \end{pmatrix} = a^*c + b^*d$$



Probabilities: How to calculate $p(D1)$ and $p(D2)$

$$|\text{photon}\rangle_{t_2} = a|\text{right}\rangle + b|\text{down}\rangle$$

$$p(D1) = |a|^2 \quad ; \quad p(D2) = |b|^2$$

Dirac notation

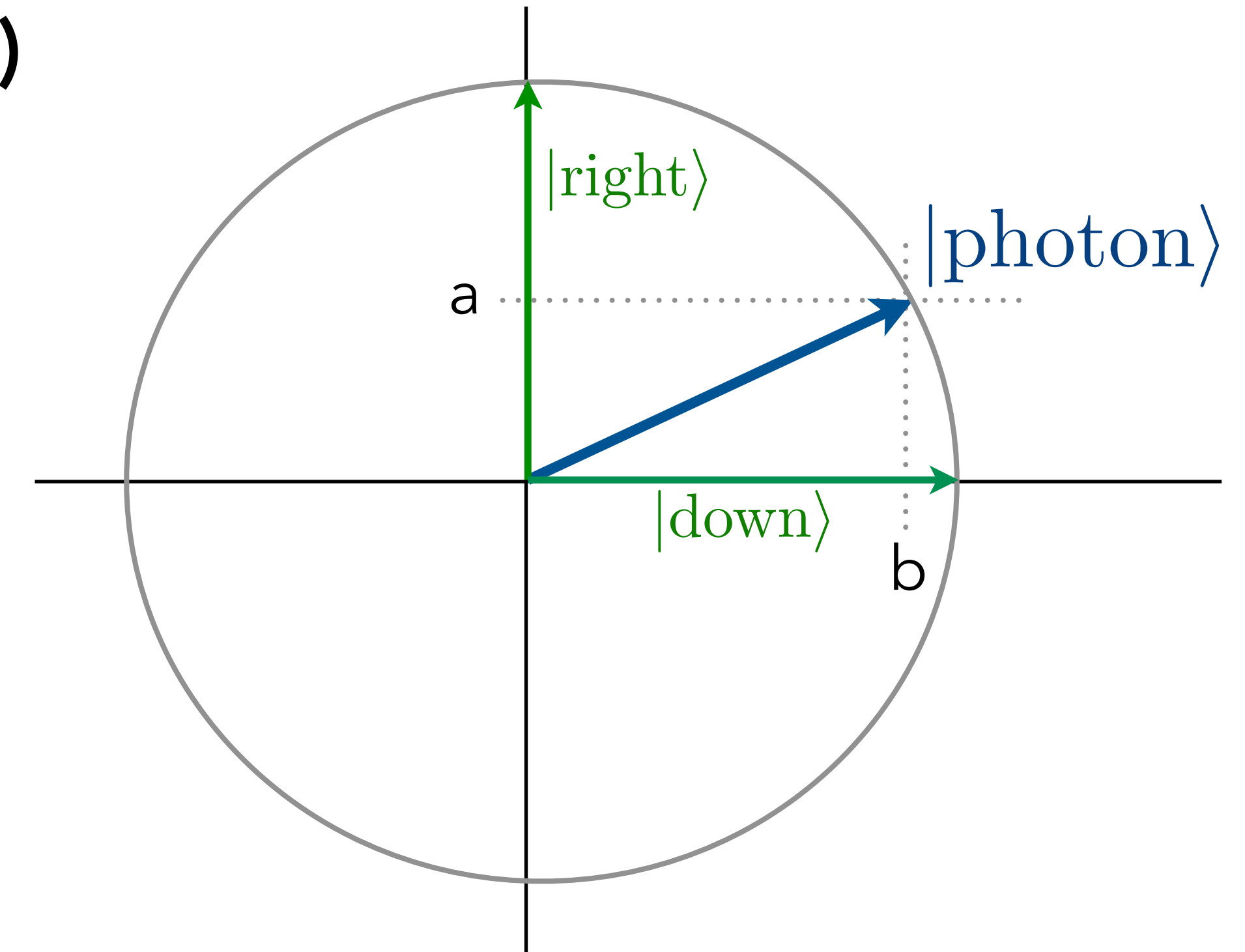
$|\dots\rangle$ ket ← vector

$\langle\dots|$ bra ← complex conjugate

$\langle\dots|\dots\rangle$ bra-ket ← scalar product

$$\langle\text{right}|\text{down}\rangle = 0 \qquad \langle\text{down}|\text{down}\rangle = 1$$

$$\langle\text{down}|\text{right}\rangle = 0 \qquad \langle\text{right}|\text{right}\rangle = 1$$



$$\langle\text{right}|\text{photon}\rangle = a\langle\text{right}|\text{right}\rangle + b\langle\text{right}|\text{down}\rangle = a$$

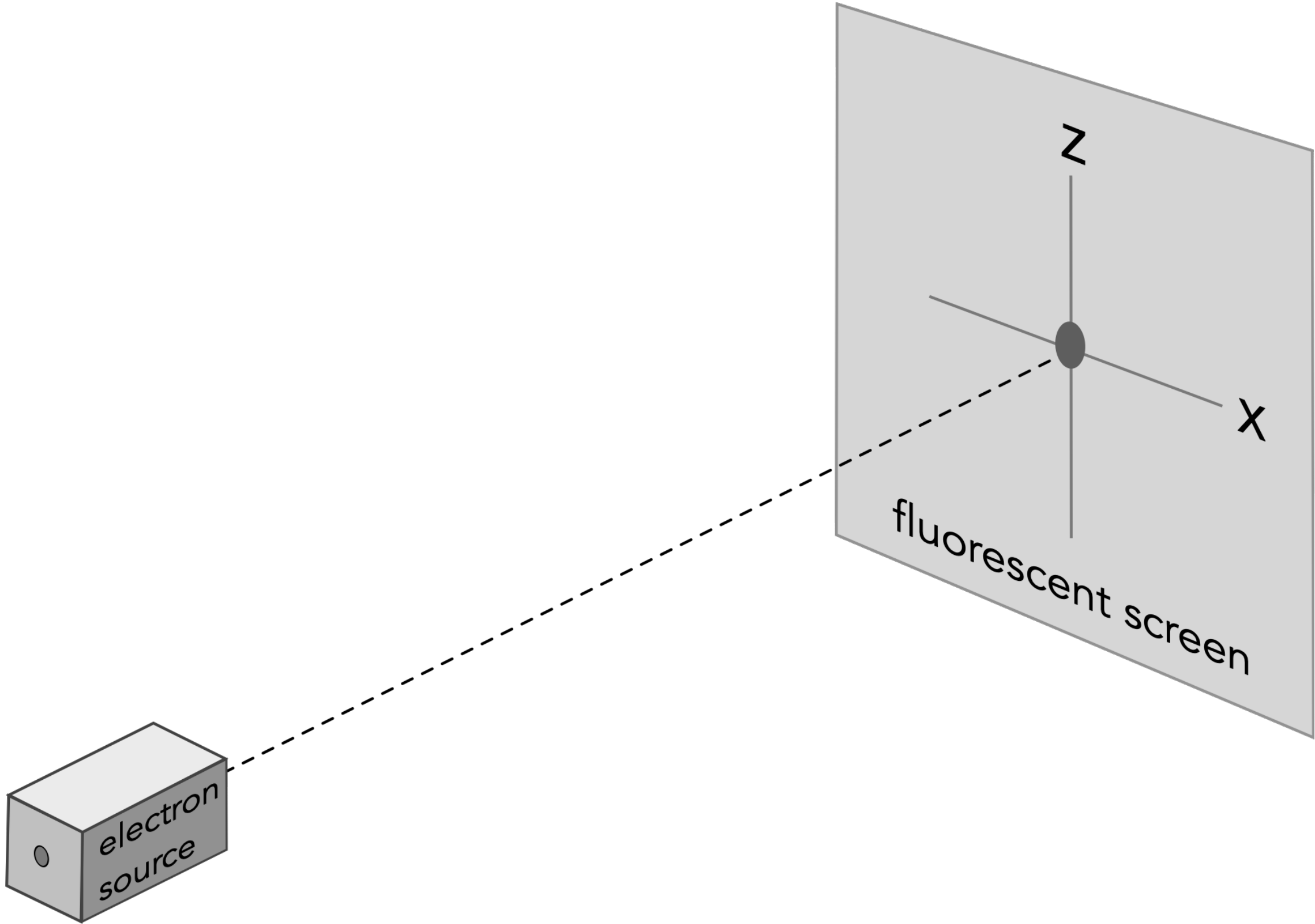
$$p(D1) = |\langle\text{right}|\text{photon}\rangle|^2 = |a|^2$$

$$\langle\text{down}|\text{photon}\rangle = a\langle\text{down}|\text{right}\rangle + b\langle\text{down}|\text{down}\rangle = b$$

$$p(D2) = |\langle\text{down}|\text{photon}\rangle|^2 = |b|^2$$

$$p(\text{outcome}) = |\langle\text{outcome}|\text{state}\rangle|^2$$

Stern-Gerlach



Stern-Gerlach

- $p(\text{up}) = p(\text{down}) = 50\%$
- random events

$$|\text{spin}\rangle = a|\uparrow\rangle + b|\downarrow\rangle \quad |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

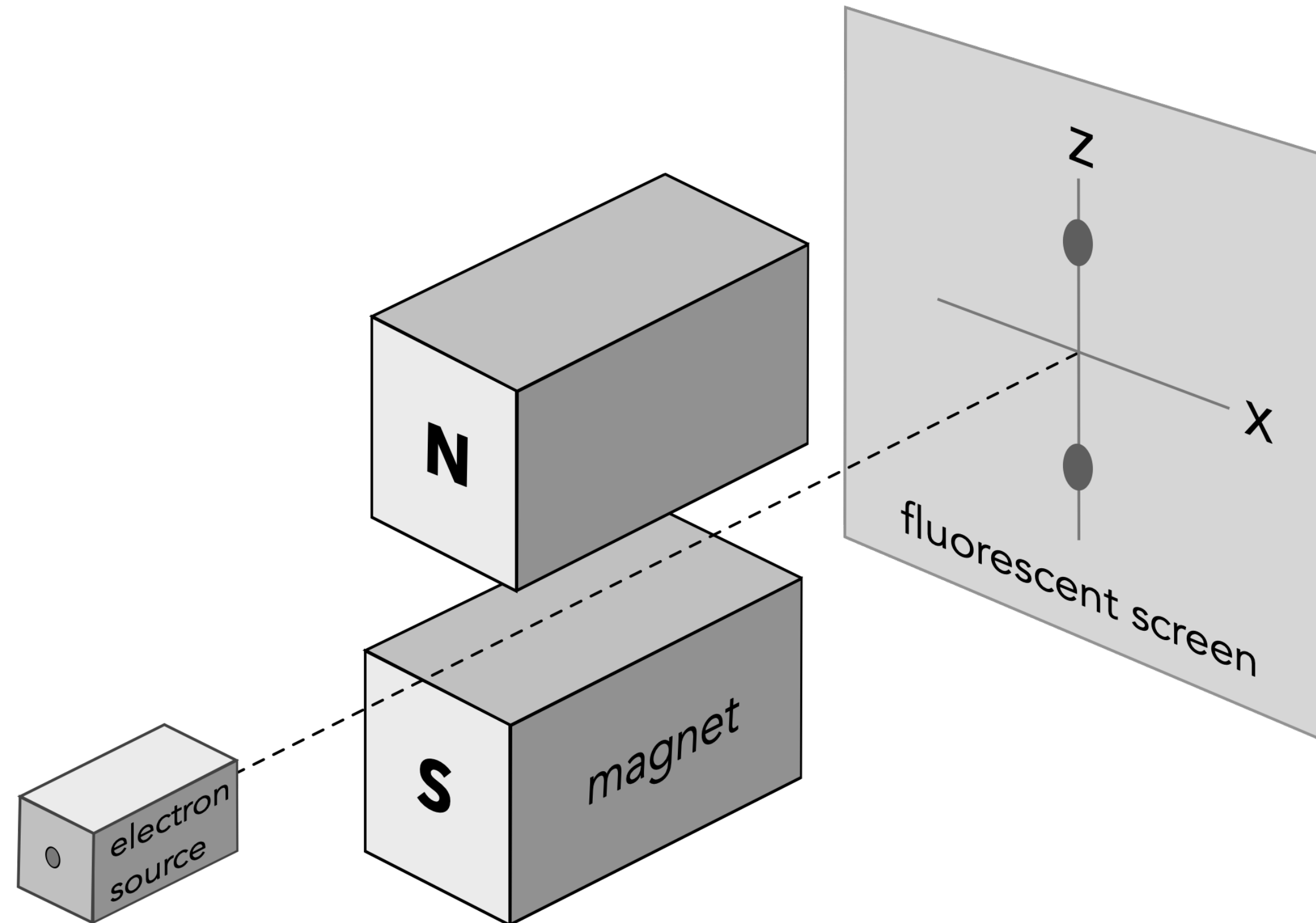
$$|\text{spin}\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\langle \text{spin} | \text{spin} \rangle = (a^* b^*) \begin{pmatrix} a \\ b \end{pmatrix} = |a|^2 + |b|^2 = 1$$

$$p_{\uparrow} = \langle \uparrow | \text{spin} \rangle = |a|^2 \quad p_{\downarrow} = \langle \downarrow | \text{spin} \rangle = |b|^2$$

$$|+\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle$$

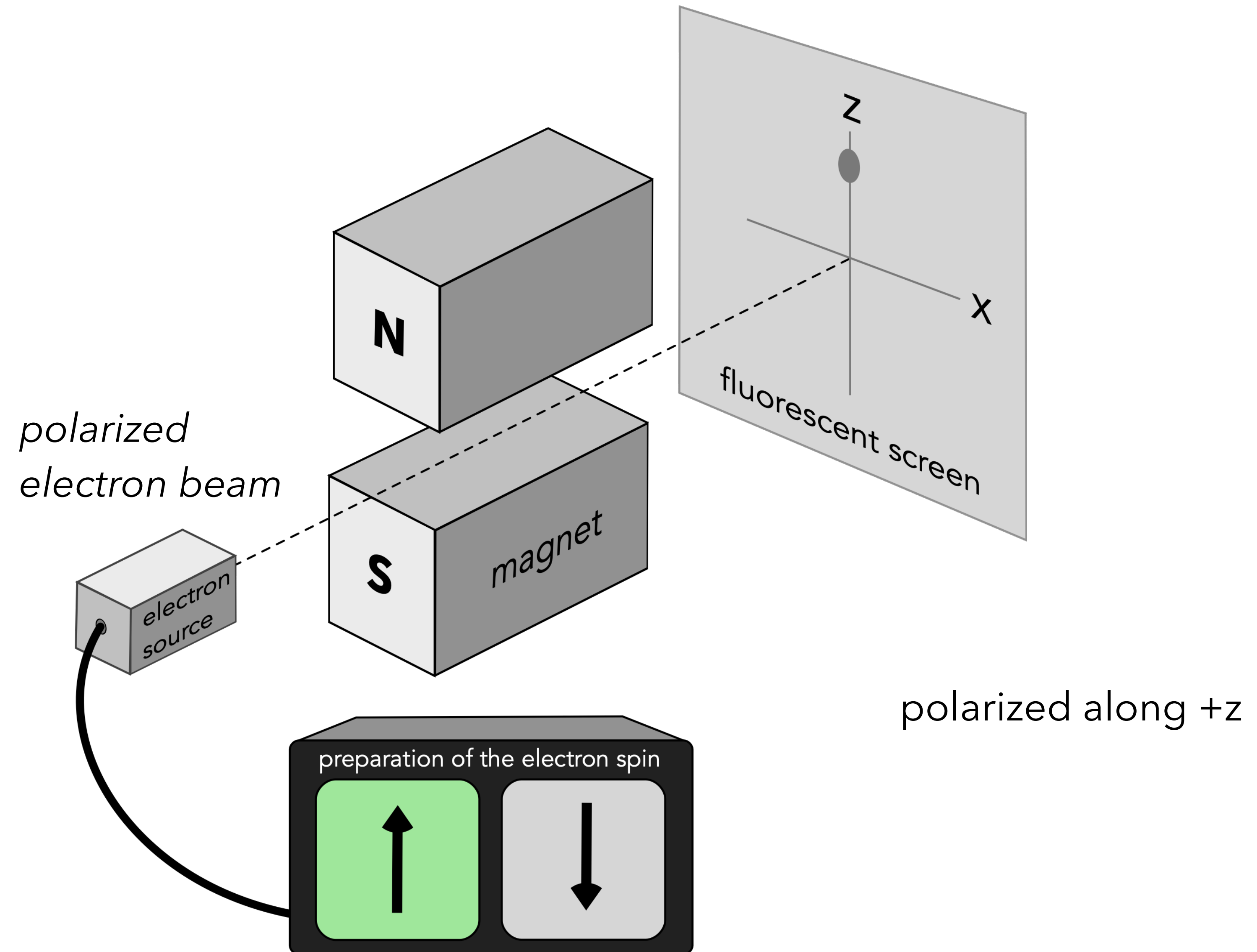
$$|-\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle$$



Stern-Gerlach

$$|\text{spin}\rangle = a|\uparrow\rangle + b|\downarrow\rangle$$

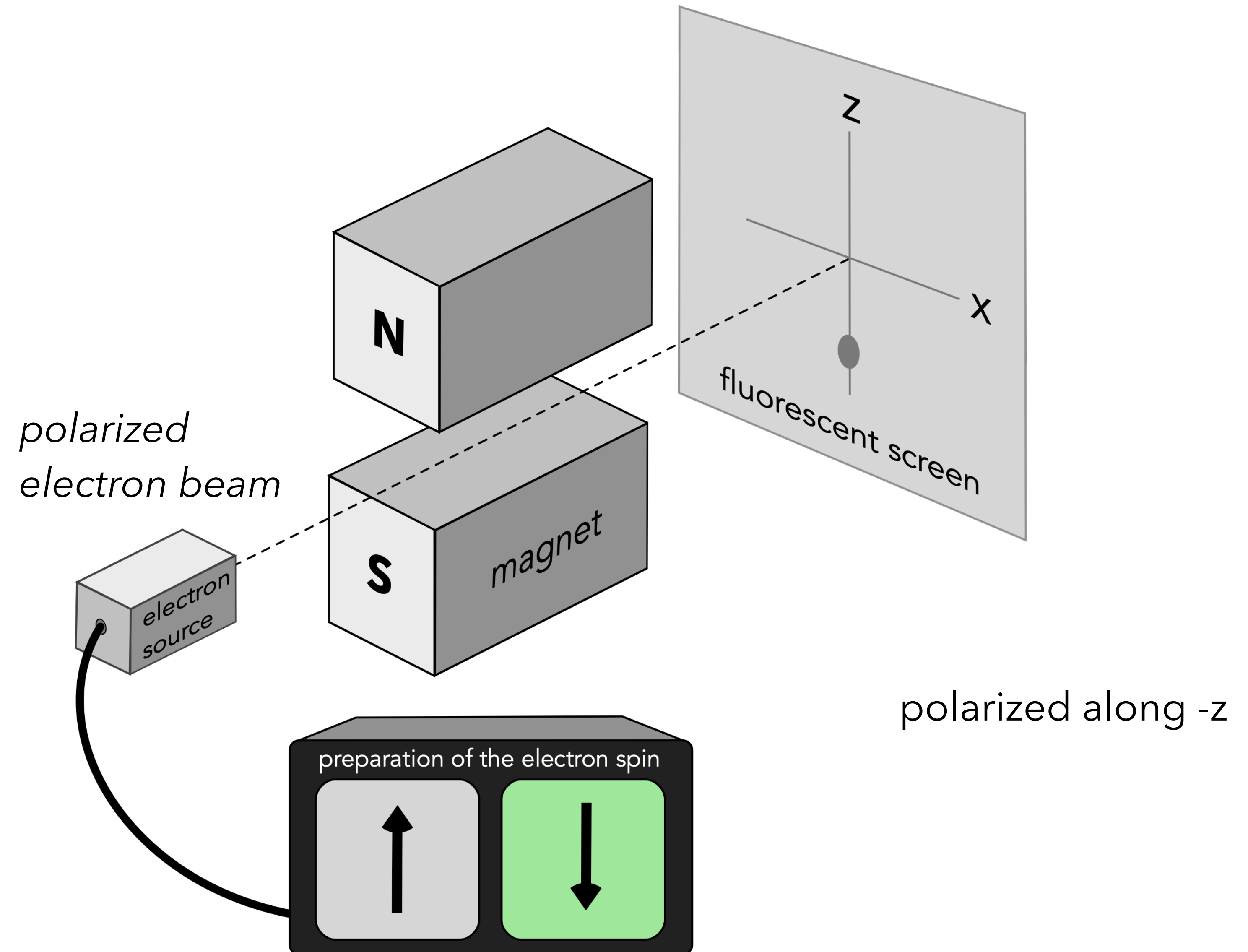
$$a = 1 ; \quad b = 0$$



Stern-Gerlach

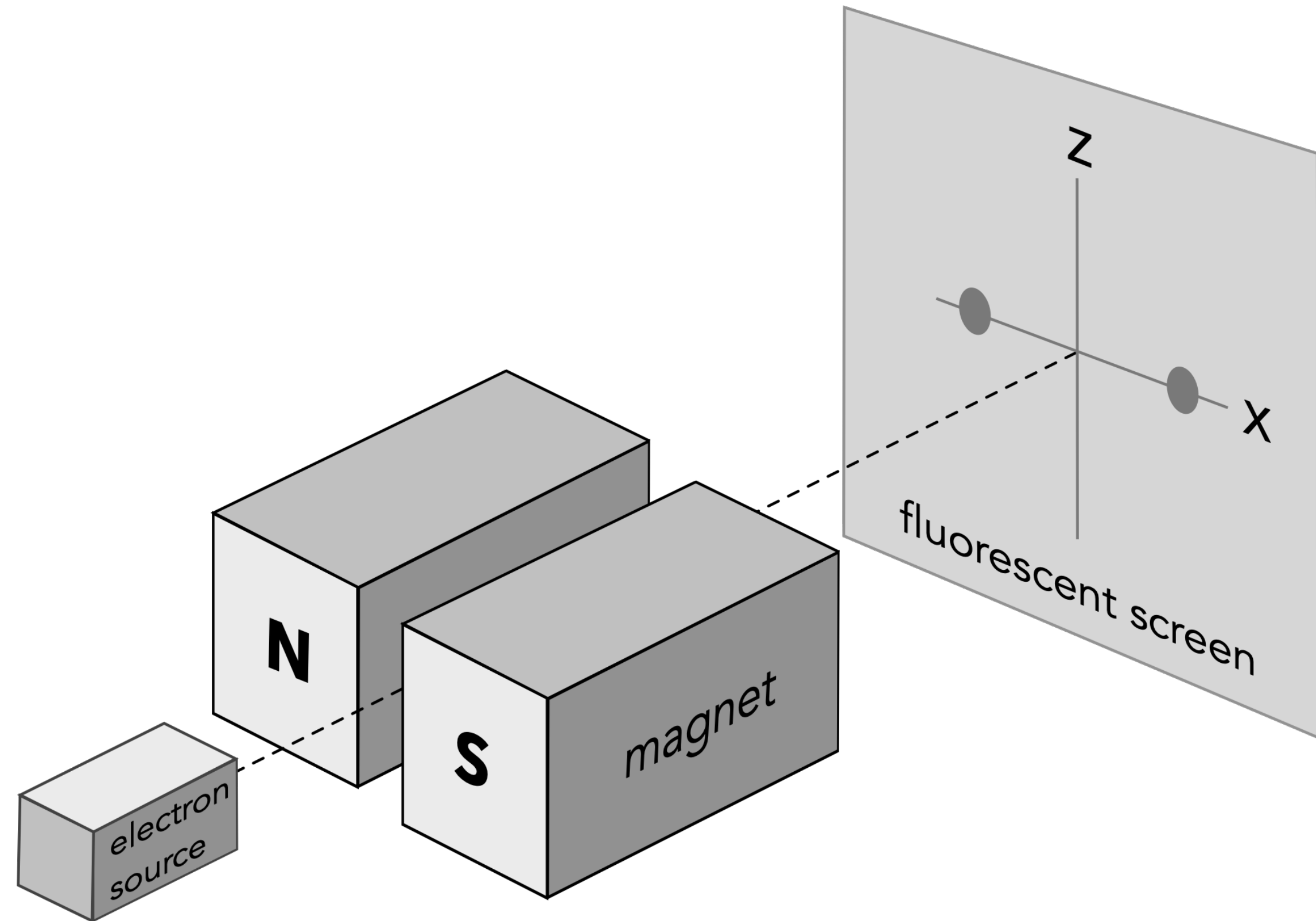
$$|\text{spin}\rangle = a|\uparrow\rangle + b|\downarrow\rangle$$

$$a = 0 ; \quad b = 1$$



Stern-Gerlach

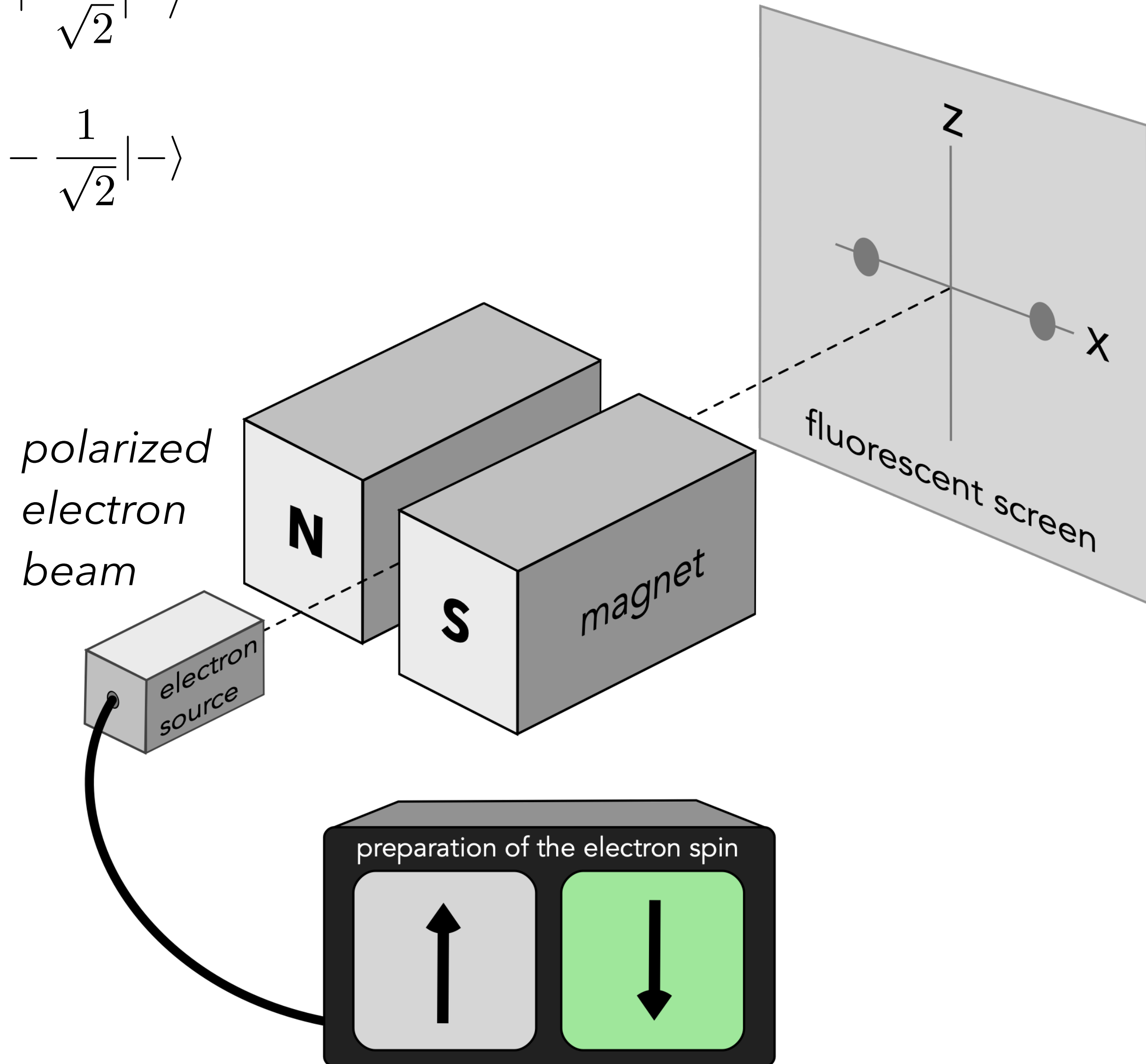
- $p(\text{left}) = p(\text{right}) = 50\%$
- random events



Stern-Gerlach

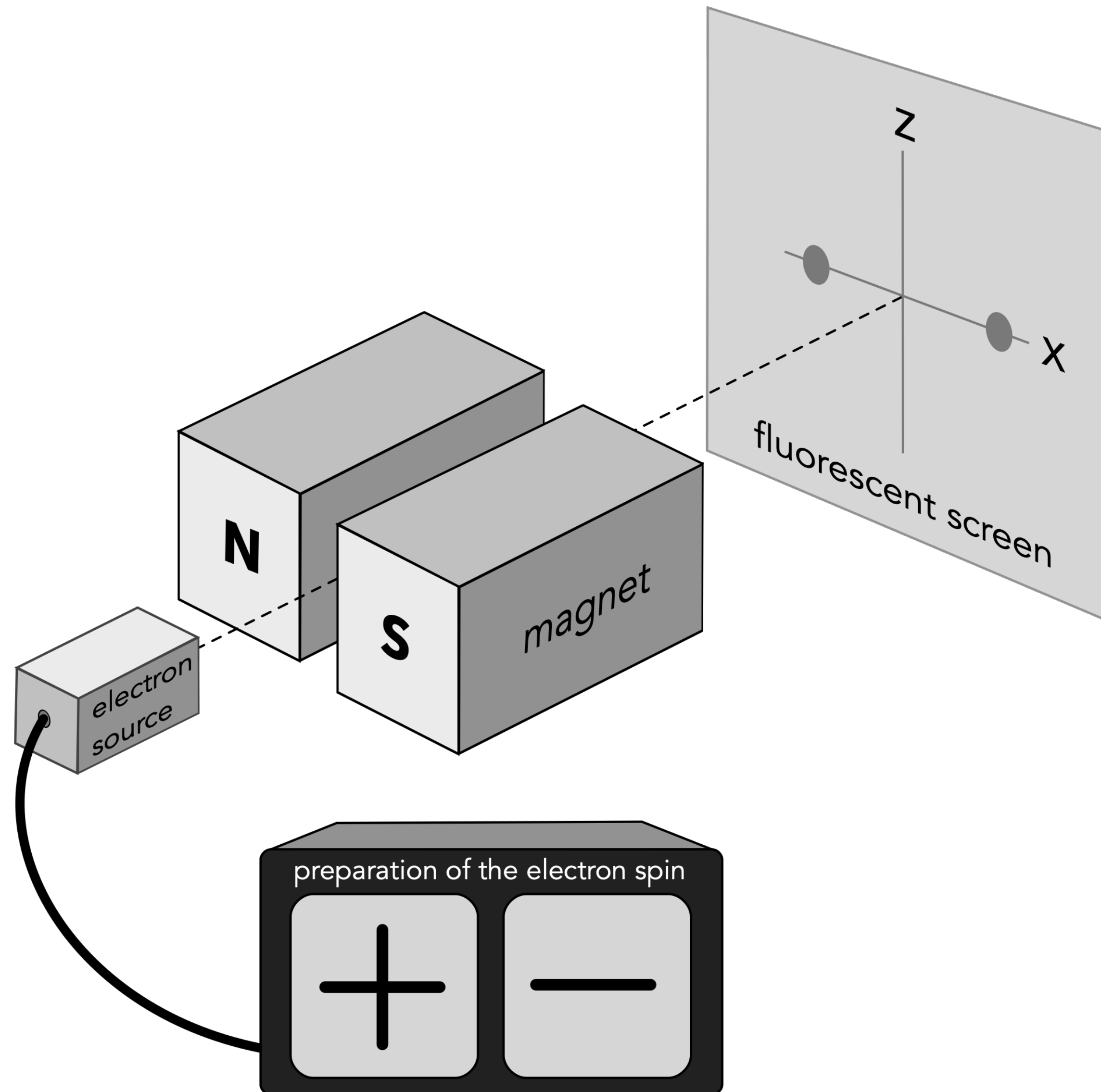
$$|+\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle \quad \Rightarrow \quad |\uparrow\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle \quad |\downarrow\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$$



Stern-Gerlach

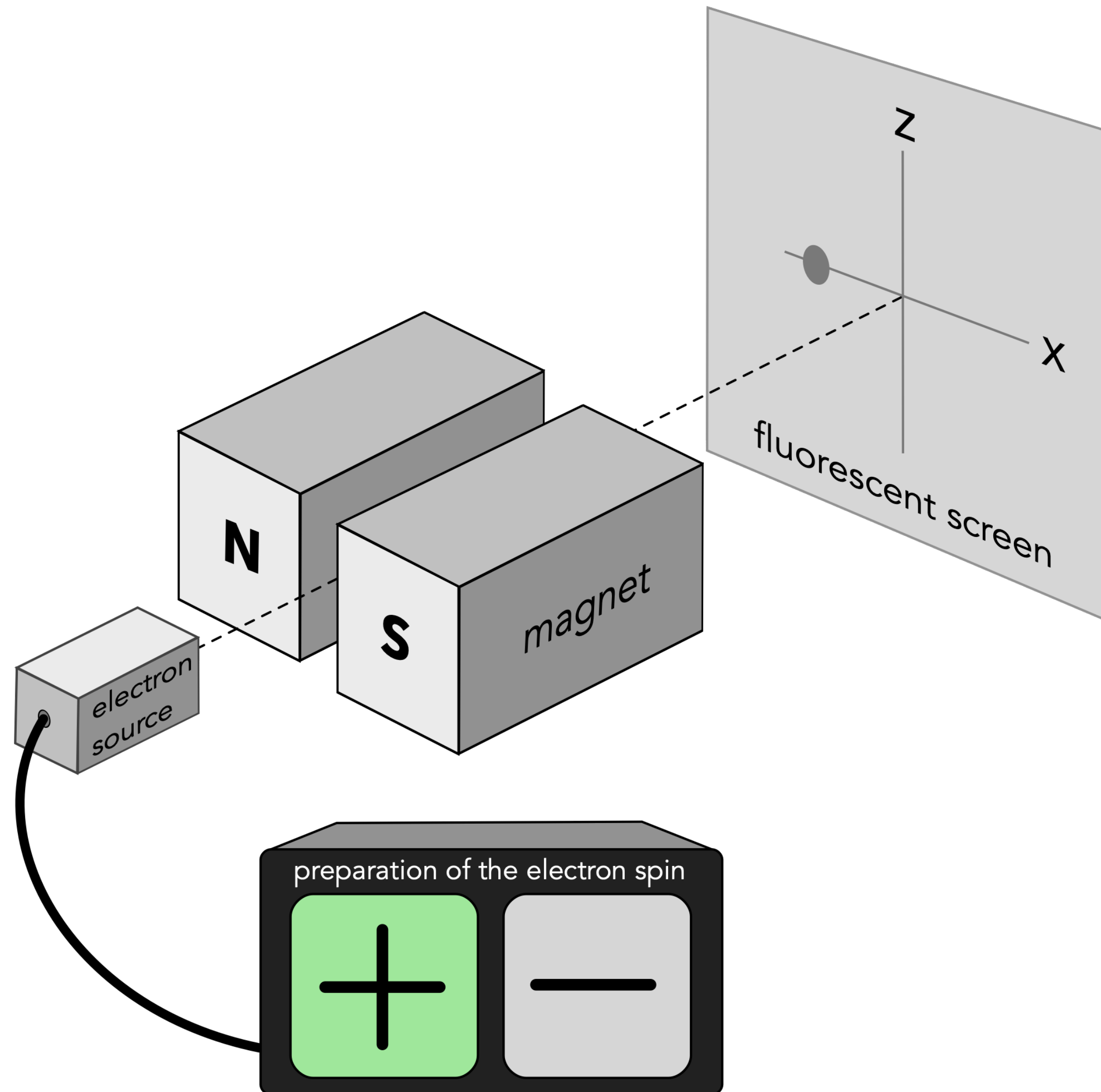
$$|\text{spin}\rangle = a|+\rangle \pm b|-\rangle$$



Stern-Gerlach

$$|\text{spin}\rangle = a|+\rangle \pm b|-\rangle$$

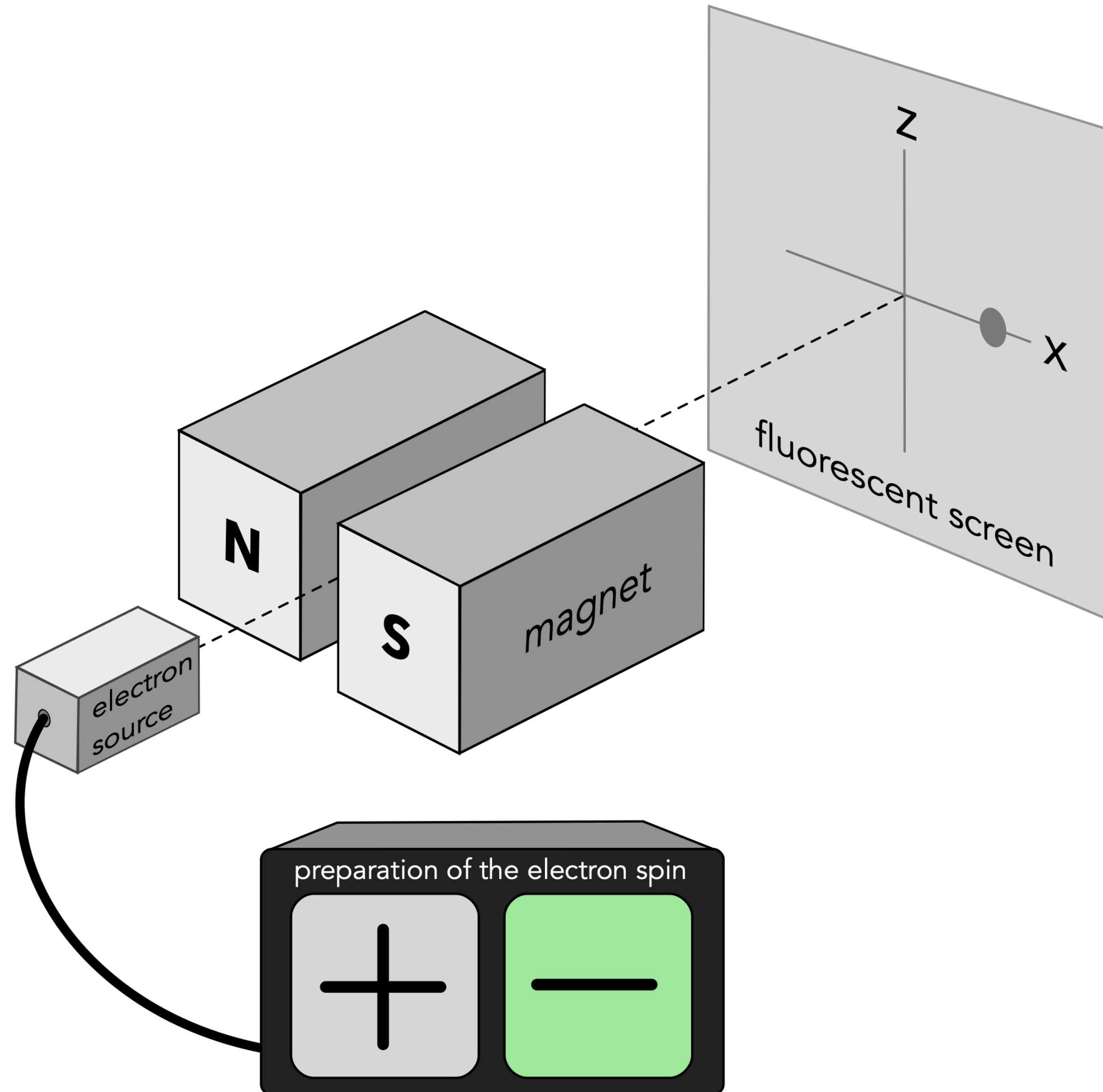
$$a = 1 ; \quad b = 0$$



Stern-Gerlach

$$|\text{spin}\rangle = a|+\rangle \pm b|-\rangle$$

$$a = 0 ; \quad b = 1$$

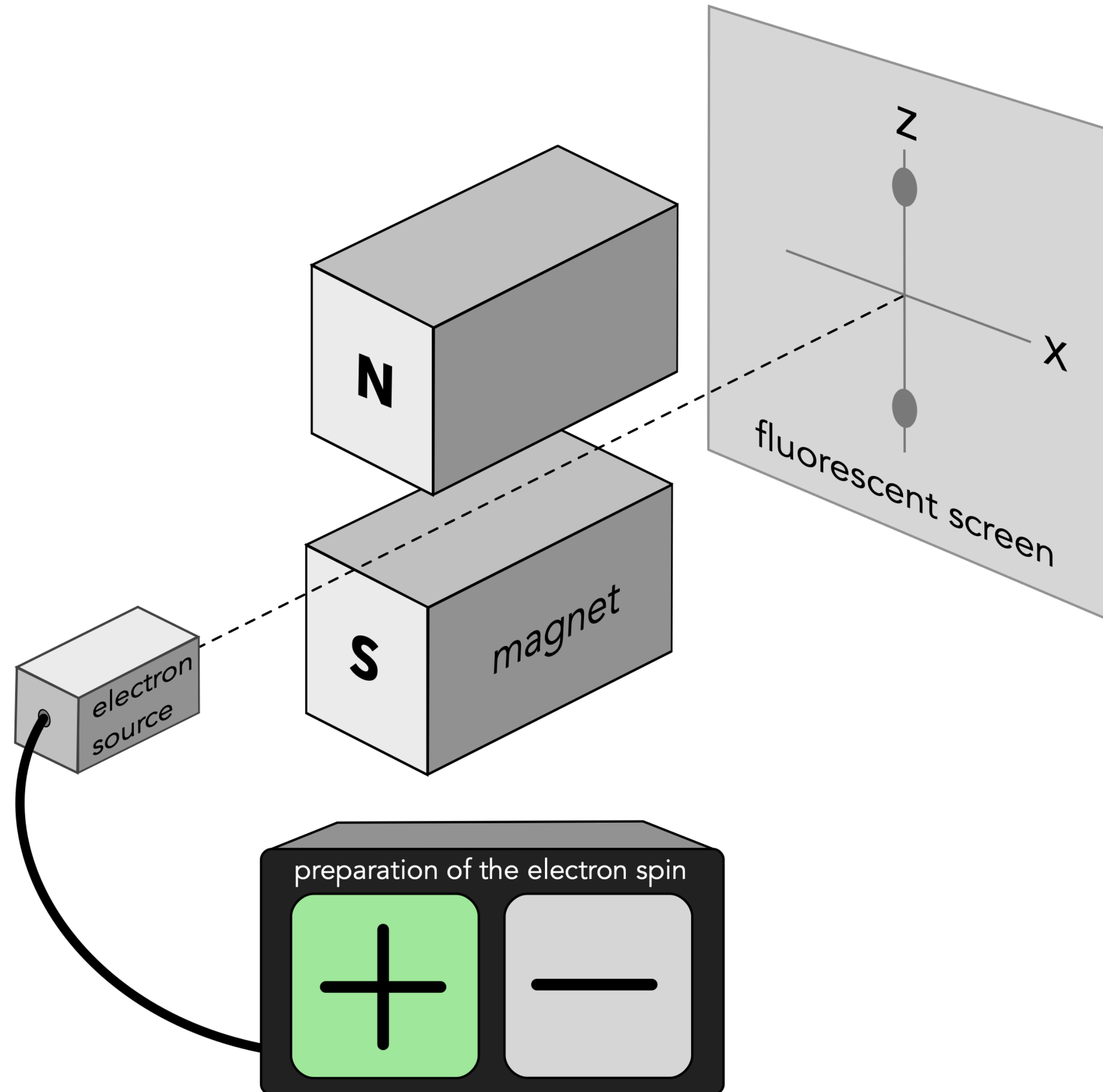


Stern-Gerlach

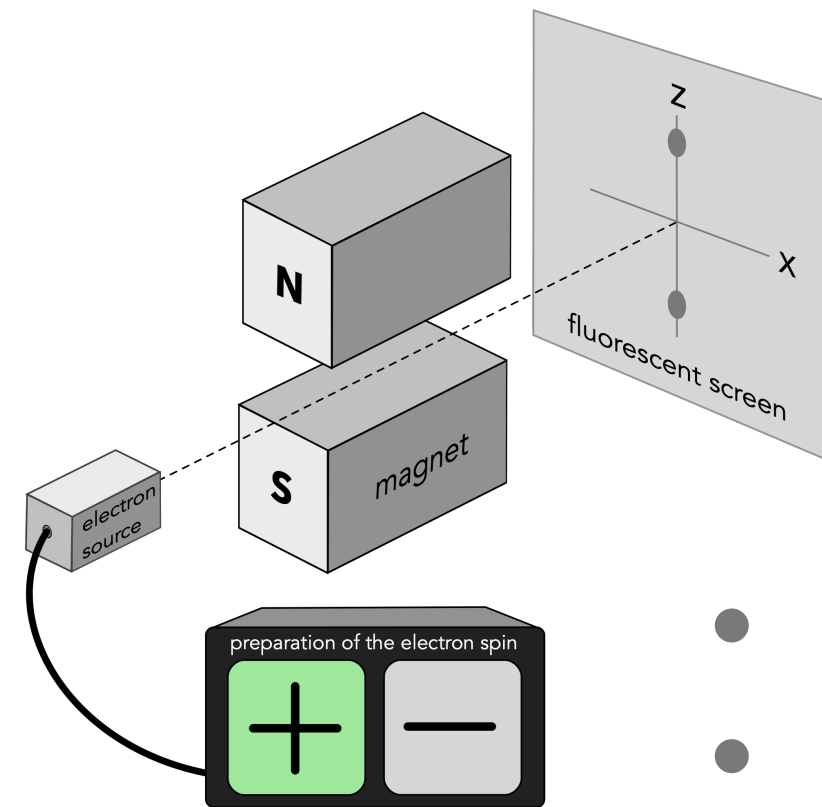
- $p(\text{up}) = p(\text{down}) = 50\%$
- random events

$$\begin{aligned} p(\uparrow) &= |\langle \uparrow | + \rangle|^2 \\ &= \left| \frac{1}{\sqrt{2}} \langle \uparrow | \uparrow \rangle + \frac{1}{\sqrt{2}} \langle \uparrow | \downarrow \rangle \right|^2 = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} p(\downarrow) &= |\langle \downarrow | + \rangle|^2 \\ &= \left| \frac{1}{\sqrt{2}} \langle \downarrow | \uparrow \rangle - \frac{1}{\sqrt{2}} \langle \downarrow | \downarrow \rangle \right|^2 = \frac{1}{2} \end{aligned}$$

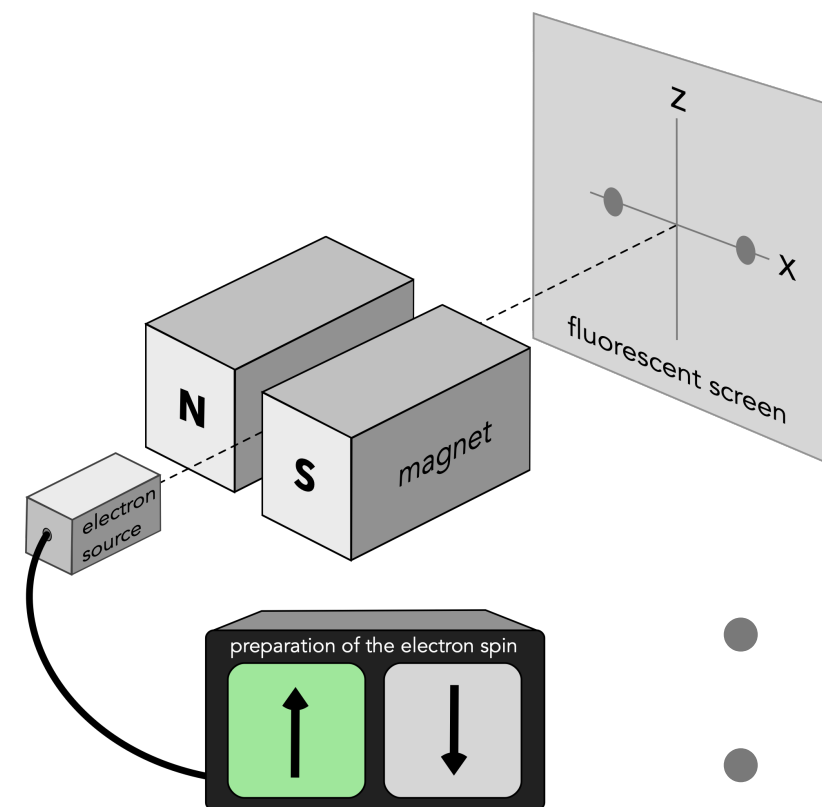


Stern-Gerlach



x polarized

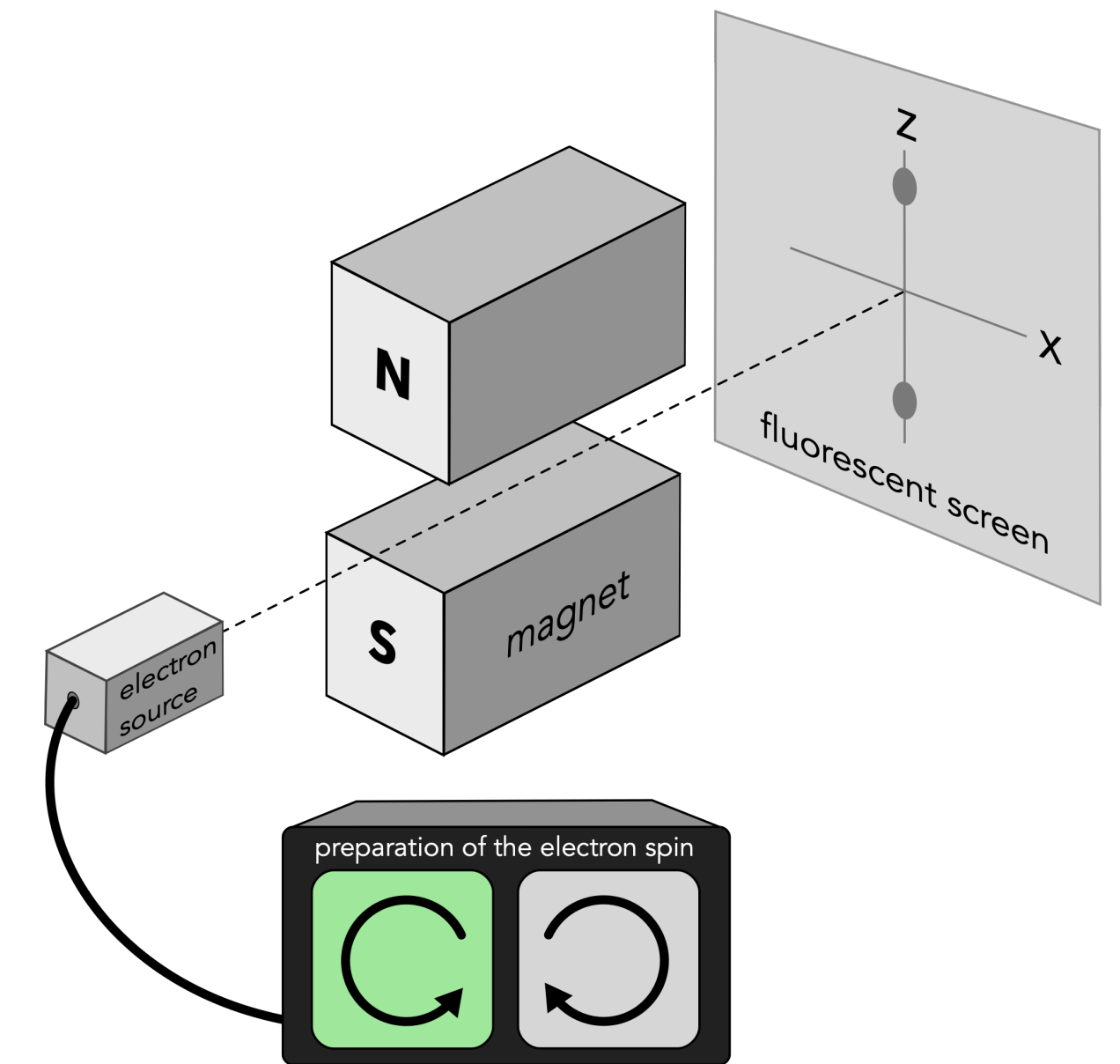
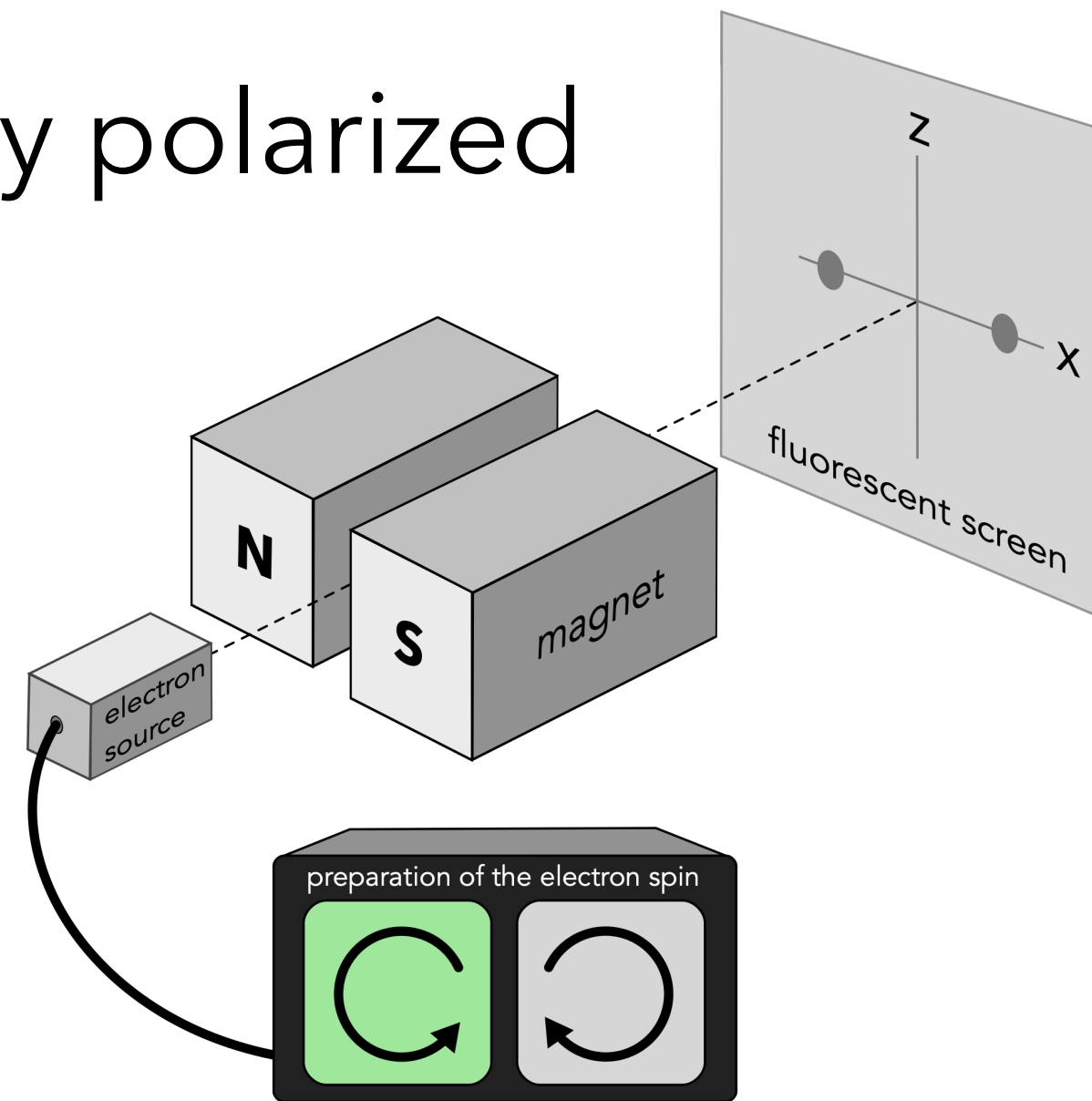
- $p(\text{up}) = p(\text{down}) = 50\%$
- random events



z polarized

- $p(\text{left}) = p(\text{right}) = 50\%$
- random events

y polarized



- $p(\text{up}) = p(\text{down}) = p(\text{left}) = p(\text{right}) = 50\%$
- random events

$$|\odot\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{i}{\sqrt{2}}|\downarrow\rangle$$

$$|\odot\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{i}{\sqrt{2}}|\downarrow\rangle$$

Stern-Gerlach

$$|\circ\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{i}{\sqrt{2}}|\downarrow\rangle$$

$$p_{\uparrow} = |\langle\uparrow|\circ\rangle|^2 = \left|\frac{1}{\sqrt{2}}\langle\uparrow|\uparrow\rangle + \frac{i}{\sqrt{2}}\langle\uparrow|\downarrow\rangle\right|^2 = \frac{1}{2}$$

$$p_{\downarrow} = |\langle\downarrow|\circ\rangle|^2 = \frac{1}{2}$$

$$p_{+} = |\langle+|\circ\rangle|^2 = \frac{1}{2}$$

$$p_{-} = |\langle-|\circ\rangle|^2 = \frac{1}{2}$$

$$|\circ\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{i}{\sqrt{2}}|\downarrow\rangle$$

$$p_{\uparrow} = |\langle\uparrow|\circ\rangle|^2 = \left|\frac{1}{\sqrt{2}}\langle\uparrow|\uparrow\rangle - \frac{i}{\sqrt{2}}\langle\uparrow|\downarrow\rangle\right|^2 = \frac{1}{2}$$

$$p_{\downarrow} = |\langle\downarrow|\circ\rangle|^2 = \frac{1}{2}$$

$$p_{+} = |\langle+|\circ\rangle|^2 = \frac{1}{2}$$

$$p_{-} = |\langle-|\circ\rangle|^2 = \frac{1}{2}$$

Spin as an observable: Expectation value

Spin is a vector

has a direction and a magnitude

is a physical observable

has a unit (Js)

is the quantum mechanical analog of the rotation around an internal axis

$$\mathbf{S} = \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix}$$

The value of the electron spin in the z-direction of $|\uparrow\rangle$ equals $\frac{1}{2}\hbar$
of $|\downarrow\rangle$ equals $-\frac{1}{2}\hbar$

$$\hbar = \frac{h}{2\pi} = 1.054 \cdot 10^{-34} \text{Js}$$

(reduced) Planck's constant

The value of the electron spin in the x-direction of $|+\rangle$ equals $\frac{1}{2}\hbar$

but what about the z-component?

Spin as an observable: Expectation value

$$|+\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle \quad \text{individual measurements of } S_z \text{ give either } \frac{1}{2}\hbar \text{ or } -\frac{1}{2}\hbar \quad \text{on average: } \langle S_z \rangle = 0$$

state with arbitrary spin $|\psi\rangle$

$$\begin{aligned}\langle S_z \rangle &= p_{\uparrow} \left(+\frac{\hbar}{2} \right) + p_{\downarrow} \left(-\frac{\hbar}{2} \right) \\ &= \frac{\hbar}{2} (p_{\uparrow} - p_{\downarrow}) \\ &= \frac{\hbar}{2} (|\langle \uparrow | \psi \rangle|^2 - |\langle \downarrow | \psi \rangle|^2) \\ &= \frac{\hbar}{2} \langle \psi | \uparrow \rangle \langle \uparrow | \psi \rangle - \frac{\hbar}{2} \langle \psi | \downarrow \rangle \langle \downarrow | \psi \rangle \\ &= \langle \psi | \left(\frac{\hbar}{2} |\uparrow\rangle \langle \uparrow| - \frac{\hbar}{2} |\downarrow\rangle \langle \downarrow| \right) | \psi \rangle \\ &= \langle \psi | \hat{S}_z | \psi \rangle \quad \text{with} \quad \hat{S}_z = \frac{\hbar}{2} |\uparrow\rangle \langle \uparrow| - \frac{\hbar}{2} |\downarrow\rangle \langle \downarrow|\end{aligned}$$

Spin as an observable: Expectation value

$$\hat{S}_z = \frac{\hbar}{2} |\uparrow\rangle\langle\uparrow| - \frac{\hbar}{2} |\downarrow\rangle\langle\downarrow|$$
$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$|\uparrow\rangle\langle\uparrow| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$
$$|\downarrow\rangle\langle\downarrow| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$
$$\Rightarrow \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Spin as an observable: Expectation value

Same story for x

$$\begin{aligned}\langle S_x \rangle &= p_+ \left(+\frac{\hbar}{2} \right) + p_- \left(-\frac{\hbar}{2} \right) = \frac{\hbar}{2} (p_+ - p_-) \\ &= \frac{\hbar}{2} (|\langle +|\psi \rangle|^2 - |\langle -|\psi \rangle|^2) = \frac{\hbar}{2} \langle \psi | + \rangle \langle + | \psi \rangle - \frac{\hbar}{2} \langle \psi | - \rangle \langle - | \psi \rangle \\ &= \langle \psi | \left(\frac{\hbar}{2} | + \rangle \langle + | - \frac{\hbar}{2} | - \rangle \langle - | \right) \psi \rangle = \langle \psi | \hat{S}_x | \psi \rangle\end{aligned}$$

$$\begin{aligned}|+\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} & |+\rangle \langle +| &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ |-\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle - |\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} & |-\rangle \langle -| &= \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\end{aligned}$$

$$\Rightarrow \hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Spin as an observable: Expectation value

and for y

$$\begin{aligned}
 \langle S_y \rangle &= p_{\uparrow} \left(+\frac{\hbar}{2} \right) + p_{\downarrow} \left(-\frac{\hbar}{2} \right) = \frac{\hbar}{2} (p_{\uparrow} - p_{\downarrow}) \\
 &= \frac{\hbar}{2} (|\langle \uparrow | \psi \rangle|^2 - |\langle \downarrow | \psi \rangle|^2) = \frac{\hbar}{2} \langle \psi | \uparrow \rangle \langle \uparrow | \psi \rangle - \frac{\hbar}{2} \langle \psi | \downarrow \rangle \langle \downarrow | \psi \rangle \\
 &= \langle \psi | \left(\frac{\hbar}{2} | \uparrow \rangle \langle \uparrow | - \frac{\hbar}{2} | \downarrow \rangle \langle \downarrow | \right) \psi \rangle = \langle \psi | \hat{S}_y | \psi \rangle
 \end{aligned}$$

$$| \uparrow \rangle = \frac{1}{\sqrt{2}} (| \uparrow \rangle + i | \downarrow \rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \qquad | \uparrow \rangle \langle \uparrow | = \frac{1}{2} \begin{pmatrix} 1 \\ i \end{pmatrix} \begin{pmatrix} 1 & -i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

$$| \downarrow \rangle = \frac{1}{\sqrt{2}} (| \uparrow \rangle - i | \downarrow \rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \qquad | \downarrow \rangle \langle \downarrow | = \frac{1}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} \begin{pmatrix} 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

$$\Rightarrow \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Spin as an observable: Expectation value and uncertainty

Pauli matrices:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Beware!

$$\hat{\sigma}_y \hat{\sigma}_z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}_y \hat{\sigma}_z - \hat{\sigma}_z \hat{\sigma}_y \neq 0$$

$$[\hat{\sigma}_x, \hat{\sigma}_z] \neq 0$$

$$[\hat{\sigma}_y, \hat{\sigma}_z] \neq 0$$

$$\hat{\sigma}_z \hat{\sigma}_y = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

$$[\hat{\sigma}_x, \hat{\sigma}_y] \neq 0$$

↘ *commutator*

Operator for the difference between one observation and the average of many: $\hat{S}_z - \langle S_z \rangle$

Expectation value: $\langle \Delta S_z \rangle = \langle \psi | \hat{S}_z | \psi \rangle - \langle S_z \rangle \langle \psi | \psi \rangle = 0$

Squared deviation: $(\Delta S_z)^2 = \hat{S}_z^2 - \langle S_z \rangle^2$

➡ Expectation value: $\langle (\Delta S_z)^2 \rangle = \langle \psi | \hat{S}_z^2 | \psi \rangle - \langle \psi | \hat{S}_z | \psi \rangle^2$

Spin as an observable: Expectation value and uncertainty

Expectation value: $\langle (\Delta S_z)^2 \rangle = \langle \psi | \hat{S}_z^2 | \psi \rangle - \langle \psi | \hat{S}_z | \psi \rangle^2$

$$|\text{spin}\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \langle \hat{S}_z \rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \Rightarrow \langle \hat{S}_z \rangle^2 = \frac{\hbar^2}{4}$$

$$\hat{S}_z^2 = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\langle \hat{S}_z^2 \rangle = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar^2}{4}$$

 variance of S_z : $\langle \hat{S}_z^2 \rangle - \langle \hat{S}_z \rangle^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{4} = 0$

Same same quantum state, other observable: $\langle (\Delta S_x)^2 \rangle = \langle \uparrow | \hat{S}_x^2 | \uparrow \rangle - \langle \uparrow | \hat{S}_x | \uparrow \rangle^2$

$$\langle \hat{S}_x \rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 \Rightarrow \langle \hat{S}_x \rangle^2 = 0$$

$$\hat{S}_x^2 = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \langle \hat{S}_x^2 \rangle = \frac{\hbar^2}{4}$$

 variance of S_x : $\langle \hat{S}_x^2 \rangle - \langle \hat{S}_x \rangle^2 = \frac{\hbar^2}{4} - 0 = \frac{\hbar^2}{4}$

Summary

$$\{| \text{top} \rangle, | \text{left} \rangle, | \text{down} \rangle, | \text{right} \rangle\}$$

$$\{| \uparrow \rangle, | \downarrow \rangle\}$$

- Abstract vector spaces of quantum states
- Linear combinations are also quantum states
- Column vectors

$$\{\langle \text{top}|, \langle \text{left}|, \dots\}$$

- Dual vector space
- Row vectors (transposed complex conjugated)

$$|\langle \psi | \phi \rangle|^2$$

- Probability of finding quantum state ψ for a system in state ϕ
- A number between 0 and 1

$$\hat{U}|\psi\rangle = |\phi\rangle$$

- Operator to transform quantum state ψ into ϕ
- N x N matrix

$$\hat{P} = |\psi\rangle\langle\psi|$$

- Operator for measuring quantum state ψ
- N x N matrix

$$\langle \phi | \hat{O} | \phi \rangle$$

- Expectation value; average value of the observable O for a system in state ϕ